



GumBall6900

A signal-directed onchain portfolio acquisition and redemption protocol

The complete technical specification: exact formulas, state-transition tables, accounting identities, security invariants, precision analysis, threat model, and verification evidence.

VERSION

2.1.0

DATE

2026-08-26

SOURCE COMMIT

uncommitted-...

PROTOCOL STATUS

Development candidate implementing ADRs through ADR 0051; not approved for user funds.

DEPLOYMENT

Not deployed on any network. No signed deployment manifest exists.

INDEPENDENT AUDIT

V12 export received for the pinned commit; incomplete assurance package, three behaviors confirmed, no release approval.

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Safety status. The protocol is **not deployed or approved for user funds**. A V12 finding export targets the exact source at `3ae171b`, but it is incomplete and not release-authorizing; independent disposition confirmed three behaviors requiring treatment. It does not cover ADR 0051's later API, batching, read periphery, SDK, or subgraph changes. Local green checks and the export remain bounded evidence, never a safety or release claim.

Governance is unselected. The core contains no `ProtocolGovernor`, `TimelockController`, generic executor, or provider-specific governance adapter; [ADR 0034](#) removed them. `Resonance` is the only core contract with continuing custom owner authority. `SignalGBX`, `StrategyFactory`, and `BribeFactory` retain setup-only Ownable shells until production explicitly renounces them after their one-time bindings are consumed. The external governance system that will own `Resonance` has not been selected. §15 and §27 specify exactly what the core does and does not guarantee as a consequence.

Companion sources. The compact whitepaper and one-page sheet are built from `docs/whitepaper/` and `docs/one-pager/gumball6900/` via `pnpm docs:whitepaper` and `pnpm docs:one-pager`. This long-form source builds to `output/pdf/GumBall6900-whitepaper.pdf` via `pnpm docs:longform`.

1. Abstract

GumBall6900 is an immutable, governance-minimized protocol that converts recurring onchain revenue into a permissionlessly redeemable portfolio of arbitrary ERC-20 assets, with allocation directed by non-transferable signal weight rather than by a manager, an oracle, or an index methodology.

The protocol issues a token, **GBX**, through a multislot mining market in which new slot tenures begin through hourly descending-price auctions denominated in an external stablecoin, **USDG**. Eighty percent of a replacement payment becomes a pull claim for the outgoing tenure miner; the remainder constitutes protocol revenue. GBX starts at zero supply and Mine is its sole lifetime issuer. A reviewed, externally created fungible Uniswap v2-style USDG/GBX LP token may be one ordinary bootstrap Strategy asset, but the core contains no liquidity-specific mechanism or guarantee.

Revenue is placed into a single rolling seven-day Synthetix-shaped emission schedule held by **Resonance** and allocated continuously across **Strategies** in proportion to the non-transferable signal weight (**SignalGBX**, ticker `sGBX`) allocated to each Strategy during each elapsed interval. A Strategy is a bounded descending-price auction that exchanges its accumulated USDG for a fixed target asset. For each purchase, Strategy snapshots a bounded global signaler rate, floors that purchase's Bribe share, transfers the complement directly to **Fund**, and sends only the Bribe share to a small notification buffer. The rate defaults to 10% and is capped at 20%; at least 80% of each individual purchase therefore reaches Fund. There is deliberately no cumulative split carry: partitioning value across purchases can change aggregate classification by sub-token rounding units (ADR 0047).

GBX holders redeem by burning GBX and nominating an arbitrary set of unique non-GBX token addresses, receiving for each the floored pro-rata share of Fund's balance against a single effective pre-burn supply snapshot that includes all accrued unminted mining. Signaler compensation has two sources: the bounded automatic acquisition share (10% by default and prospectively adjustable from 0% through 20%), and **Bribes**, permissionlessly funded reward streams attached to each Strategy, capped at sixteen reward tokens.

Protocol administration is reduced to four continuing owner-gated calls on a single contract: `Resonance.addStrategy`, `Resonance.killStrategy`, `Resonance.addBribeRewardToken`, and bounded prospective `Resonance.setBribeBps`. `SignalGBX`, `StrategyFactory`, and `BribeFactory` retain setup-only Ownable shells after their one-time bindings; production must remove those temporary owners. The remaining core contracts are ownerless or authorized by immutable contract identity rather than an owner. The core implements no proposal, quorum, voting, delay, or cancellation semantics of its own; ADR 0034 removed the in-repository Governor and Timelock in favour of an external governance system that has not yet been selected. `SignalGBX` retains non-transferable ERC20Votes checkpoints for that future integration, and the

core assigns them no meaning. Selection, review, and the ownership handoff that removes the temporary deployment owner remain deployment blockers.

This document specifies the implemented mathematics exactly, states the accounting identities the implementation can actually prove (and explicitly declines to assert those it cannot), enumerates state transitions, and presents the threat model and residual risks.

2. Motivation

Pooled onchain asset vehicles have converged on a small set of trust concessions that are, individually, defensible and collectively fatal to the claim of trust minimization:

1. **Discretionary allocation.** A manager, multisig, or unbounded DAO decides what the vehicle holds. Holders trust both competence and honesty.
2. **Upgradeability.** A proxy admin can change redemption arithmetic after deposits are taken.
3. **Pausability.** An emergency actor can suspend exit while entry or accrual continues.
4. **Oracle dependence.** A price feed determines mint and redemption ratios, importing the oracle's failure modes and manipulation surface into the vehicle's core accounting.
5. **Registry curation.** An asset allow-list is maintained by a privileged party, so a single broken or malicious entry can block redemption for every other asset.

Each concession exists for a reason. Removing them requires replacing the function they served with a mechanism that does not require trust.

GumBall6900 makes five substitutions:

CONCESSION REMOVED	REPLACEMENT MECHANISM
Discretionary allocation	Continuous, revocable, per-Strategy signal allocation that directs revenue by weight
Upgradeability	Immutable, non-proxied deployment with reciprocal one-time bindings
Pausability	Unpausable redemption; caller-selected redemption baskets and scalar reward claims isolate broken assets
Oracle pricing	Bounded descending-price auctions in which the clearing fill <i>is</i> price discovery
Registry curation	Registry-free raw-token treasury; the redeemer nominates the assets they wish to receive

The design accepts specific, enumerated costs for these substitutions — permanent dust accumulation, unrecoverable deployment errors, unbounded reward abandonment in retired Strategies, and the absence of any emergency response. These are stated in §38 and §39 rather than minimized.

3. Design goals

- **G1 — Immutability.** No contract may be upgraded, paused, migrated, or administratively drained. When a design choice trades governance flexibility against immutability, immutability wins and the consequence is recorded.
- **G2 — Minimal continuing authority.** The permanent administrative surface must be enumerable, selector-bounded, and small enough to state in one sentence.
- **G3 — Oracle independence.** No protocol accounting may depend on an external price, NAV, or valuation.
- **G4 — Exit liveness.** Redemption and signal removal must never depend on the cooperation, solvency, or correctness of any third-party token other than the one being withdrawn.
- **G5 — Failure isolation.** Where the core offers an isolated operation, a malformed or frozen token should block only that operation: scalar Bribe claims isolate reward tokens, and caller-selected Fund baskets isolate assets.

- **G6 — Standard-token boundary.** Canonical GBX/USDG, Strategy payment tokens, and Bribe reward tokens are assumed to be standard, non-rebasing ERC-20s. `SafeERC20` handles return-value compatibility; Mine, SignalGBX, reward, and settlement paths do not duplicate balance snapshots around transfers. Fund preserves stricter checks for caller-selected arbitrary assets.
- **G7 — Bounded work.** Every mandatory loop — reward tokens, mining slots, redemption baskets — must be bounded by a code constant or by caller-supplied input the caller pays for.
- **G8 — No retroactive redirection.** A weight change must never redirect value that accrued under prior weights.
- **G9 — Explicit accounting.** Where exact conservation is achievable it must be proven by identity; where it is not achievable, the residue must be classified and disclosed rather than implied to be zero.

4. Non-goals

- **N1 — No index methodology.** The set of Strategies registered in `Resonance` constitutes the protocol's index membership: it is the curated list of assets the protocol targets. What the protocol does **not** provide is index methodology — there is no target weighting, rebalancing, drift correction, reconstitution rule, or NAV. Fund holds whatever Strategies acquired plus whatever anyone sent it, in whatever proportions resulted. Membership is defined by registration in `Resonance`, never by a Fund balance (§24.2).
- **N2 — Not a valuation system.** The protocol never computes NAV, backing-per-token, or asset prices onchain.
- **N3 — Not a yield product.** No return, distribution, or performance is promised or engineered.
- **N4 — Not a general DAO.** The core implements no voting, proposal, or execution machinery at all. Protocol administration is four continuing owner-gated calls on one contract (§15.1).
- **N5 — Not fee-on-transfer or rebase compatible.** Those mechanics are outside the supported-token model and may revert, underfund, or otherwise make the affected market unusable; registration is a governance responsibility.
- **N6 — No emergency response.** There is deliberately no guardian, veto, circuit breaker, or recovery path.
- **N7 — No keeper infrastructure.** No function pays a caller bounty. All maintenance is voluntary and permissionless.
- **N8 — No reward-dust conservation guarantee.** `Resonance` and `Bribe` both accept rate-, index-, and account-level floors as unallocated surplus. No lifetime bound on that residue is claimed.

5. Terminology

TERM	DEFINITION
GBX	Transferable ERC-20 with ERC-2612 permit, 18 decimals. No vote checkpoints. Mined, escrowed through SignalGBX, burned.
sGBX / SignalGBX	Non-transferable ERC-20 with ERC20Votes, 18 decimals. Minted 1:1 against GBX atomically assigned to Strategy signal.
USDG	External stablecoin used for revenue. Six decimals by deployment assumption , not by code enforcement.
Signal	An absolute quantity of sGBX a specific account has allocated to a specific Strategy.
Account aggregate signal	The account's sGBX balance, equal to its allocations across all live and killed Strategies; returned as GBX only by removing a paired allocation.
Strategy	A bounded descending-price auction exchanging accumulated USDG for one fixed payment token.
Live / killed Strategy	A Strategy accepting new signal and future revenue / one permanently excluded from both.
Bribe	Per-Strategy multi-token reward stream, permissionlessly funded, paid to that Strategy's signalers.
BribeRouter	Per-Strategy Bribe-only buffer that permissionlessly notifies its complete qualifying payment-token balance.

TERM	DEFINITION
Fund	Ownerless raw-token treasury. Redemption and GBX burning are its only value exits.
Slot	One mining position accruing GBX at a tenure-locked rate; each new tenure begins through an hourly auction.
Epoch	One auction round, identified by a monotonically increasing <code>epochId</code> used for fill-race protection.
Reverse Dutch auction	The repository's term for the descending-price mechanism in <code>Mine</code> and <code>Strategy</code> . See §43 discrepancy D-1.
Reward period / stream	A seven-day whole-unit-per-second schedule; valid active top-ups roll the ordinary scheduled leftover into a restarted period.
Surplus	Value held by a contract that is not a liability to anyone and has no recovery path.
Checkpoint	Advancing lazily-accrued state to the current timestamp before mutating weights or balances.
Resonance owner	The single address holding the four continuing administration capabilities; intended to become an external governance executor (§15).

5.1 Notation

Throughout, $\lfloor x \rfloor$ denotes integer floor division as performed by the EVM. All quantities are raw integer token units unless explicitly stated otherwise. $1 \text{ GBX} = 10^{18}$ raw units. Under the intended deployment, $1 \text{ USDG} = 10^6$ raw units. Timestamps are `block.timestamp` in seconds. `mulDiv(a, b, c)` denotes OpenZeppelin's `Math.mulDiv`, which computes $\lfloor a \cdot b / c \rfloor$ at 512-bit intermediate precision and therefore cannot overflow on the intermediate product.

6. Actors

ACTOR	CAPABILITY	TRUSTED FOR
Miner	Pays USDG to begin a slot tenure; accrues GBX at a tenure-locked rate; claims outgoing-tenure payments.	Nothing
Signaler / voter	Atomically deposits GBX, receives non-transferable voting sGBX, and assigns every unit to one live <code>Strategy</code> .	Nothing
Strategy buyer	Fills a <code>Strategy</code> auction, paying the target asset and receiving accumulated USDG.	Nothing
Bribe funder	Permissionlessly streams reward tokens to a <code>Strategy</code> 's signalers.	Nothing
Redeemer	Burns GBX and nominates assets to withdraw pro rata from Fund.	Nothing
Checkpointner / router	Permissionlessly advances lazy state and forwards qualifying revenue.	Liveness only
Resonance owner	Adds <code>Strategies</code> , kills <code>Strategies</code> , registers Bribe reward tokens, and sets the bounded prospective Bribe share (§15.1).	Fully trusted for those four capabilities
Deployment coordinator	Executes one-time bindings, creates bootstrap <code>Strategies</code> , transfers Resonance, and removes the three setup-shell owners.	Fully trusted, once, irreversibly

The deployment coordinator is the protocol's single unavoidable trusted party. Its authority is temporary by procedure rather than by code (§28.4), and every error it can make is permanent.

7. System overview

The protocol source defines eleven core contract types. The economic cycle has five stages.

Stage 1 — Issuance and revenue origination. `Mine` holds exactly sixteen permanent slots. Each slot continuously accrues GBX at a rate fixed for its current tenure. A new tenure begins by paying the slot's current hourly descending price

in USDG. On an occupied slot, `[price·8000/10000]` accrues as a pull claim for the outgoing tenure miner and the remainder is deposited into `ResonanceRouter`; on an empty slot the whole payment is deposited there. Mine's action ends after a successful `SafeERC20` request for the nominal deposit. Under the supported standard USDG model, that amount reached the Router. A later permissionless `route()` call is required to forward Router USDG into Resonance; no caller role, bounty, or liveness guarantee exists.

Stage 2 — Revenue scheduling. `ResonanceRouter` withholds any nonzero balance smaller than either seven days in raw units or the whole USDG returned by `remainingRevenue()` for Resonance's active period. Once its balance satisfies both gates and someone calls `route()`, it forwards the entire balance. Resonance checkpoints elapsed emission, combines the notification with the ordinary Synthetix leftover, floors the new per-second rate, and restarts a seven-day schedule.

Stage 3 — Signal-weighted allocation. `SignalGBX` is the sole external signal coordinator. Its restricted hooks on `Resonance` checkpoint elapsed emission before mutating weights. Resonance maintains a Synthetix-shaped cumulative index at `1e36` precision over `totalSignalWeight`, the sum of recorded weights across *live* Strategies only.

Stage 4 — Acquisition. `Strategy.buy` first pulls the Strategy's released USDG from Resonance, then exchanges its entire USDG balance for the current descending price in the Strategy's fixed payment token. Before touching that token, Strategy snapshots Resonance's global 0%-to-20% Bribe rate. It floors the Bribe share for this purchase, transfers the 100%-to-80% complement directly to immutable Fund, and transfers only a nonzero Bribe share to `BribeRouter`. Permissionless `BribeRouter.route()` later notifies the paired Bribe once the complete buffer satisfies the Bribe's duration and active-leftover gates.

A reviewed, externally created fungible Uniswap v2-style USDG/GBX LP token may be registered during bootstrap as an ordinary Strategy payment token. It uses this same acquisition and settlement path; no core component creates, seeds, owns, custodies, prices, rebalances, compounds, harvests, or swaps liquidity.

Stage 5 — Redemption. `Fund.redeem` reads Mine's constant-time effective supply without checkpointing or iterating the slots, snapshots its balance of each nominated token, pulls and burns the redeemer's GBX, and transfers each floored pro-rata share atomically.

Signaler compensation is delivered by `Bribe` contracts, funded both automatically by the bounded acquisition share (10% by default, prospectively adjustable from 0% through 20%) and by anyone who chooses to add further rewards.

8. Contract graph

ELEVEN DEPLOYED CONTRACT TYPES · ONE CONTINUING CUSTOM OWNER-AUTHORITY EDGE

TOKEN LAYER



mint authority, once

ISSUANCE AND REVENUE



notifyRevenue

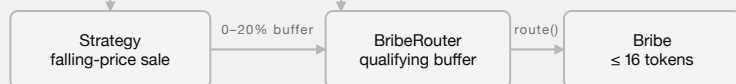
signal changes

IVOTES CHECKPOINTS ARE KEPT
BUT READ BY NOTHING IN THE CORE

ALLOCATION



PER-STRATEGY GRAPH, ONE SET EACH



80-100% direct

CUSTODY

Redemption and GBX burning
are the only exits.



owns

RESONANCE OWNER

External. Not selected.
Not in this repository.

ADDSTRATEGY · KILLSTRATEGY
ADDBRIBEREWARTOKEN
SETBRIBEBPS (0-20%)

RESONANCE HAS THE ONLY CONTINUING CUSTOM OWNER AUTHORITY. SIGNALGBX, STRATEGYFACTORY, AND BRIBEFACORY RETAIN SETUP-ONLY OWNABLE SHELLS UNTIL PRODUCTION RENOUNCES THEM. THE EXTERNAL RESONANCE OWNER IS UNSELECTED AND NOT PART OF THIS REPOSITORY.

THE DEPLOYED CONTRACT GRAPH. STRATEGY SENDS FUND'S PER-PURCHASE COMPLEMENT DIRECTLY AND ONLY THE BRIBE SHARE TO ITS QUALIFYING BUFFER. ELEVEN CONTRACT TYPES, ONE CONTINUING CUSTOM-OWNER AUTHORITY EDGE, THREE SETUP-ONLY OWNABLE SHELLS THAT PRODUCTION MUST EXPLICITLY RENOUNCES, AND AN EXTERNAL RESONANCE OWNER DELIBERATELY DRAWN OUTSIDE THE SYSTEM BECAUSE ADR 0034 REMOVED GOVERNANCE IMPLEMENTATION FROM THIS REPOSITORY.

8.1 Cardinality

CONTRACT	INSTANCES
GBX	1
SignalGBX	1
Mine	1
ResonanceRouter	1
Resonance	1
StrategyFactory	1
BribeFactory	1
Fund	1
Strategy	n, one per registered Strategy
BribeRouter	n, one per Strategy, deployed with it
Bribe	n, one per Strategy, deployed before it

9. Authority and ownership graph

NO CONTINUING CUSTOM OWNER AUTHORITY

No local custom owner-administration calls.

Mine	sixteen slots, fixed
Fund	the treasury itself
GBX	minter locked once
Strategy	auction parameters fixed
BribeRouter	Bribe-only qualifying buffer

CONTINUING CUSTOM OWNER AUTHORITY

Resonance — four custom protocol calls:

- add a Strategy
- retire a Strategy
- register a reward token
- set Bribe share (0–20%)

The holder of that address is **not yet chosen**.

There is no upgrade path, pause switch, sweep, or migration route anywhere in the protocol — so this map is the complete authority surface, not a summary of it.

SignalGBX, StrategyFactory, and BribeFactory retain setup-only Ownable shells until production explicitly renounces them; after binding, those owners have no custom protocol call.

THE COMPLETE CUSTOM PROTOCOL AUTHORITY SURFACE: FOUR BOUNDED CALLS ON RESONANCE, WITH NO LOCAL ADMINISTRATION ON THE CONTRACTS AT LEFT. INHERITED OWNERSHIP TRANSFER AND RENUNCIATION REMAIN; WHO HOLDS THAT OWNER ADDRESS IS THE LARGEST OPEN QUESTION.

9.1 Owned contracts and their permanent authority

CONTRACT	OWNER AFTER SETUP	CONTINUING OWNER-GATED FUNCTIONS
Resonance	External executor (unselected)	addStrategy , killStrategy , addBribeRewardToken , bounded setBribeBps , plus inherited transferOwnership / renounceOwnership
Mine	none	none — sixteen slots are fixed at construction
SignalGBX	(setup owner)	none remaining — setResonance is consumed at deployment
StrategyFactory	(setup owner)	none remaining — setResonance is consumed at deployment
BribeFactory	(setup owner)	none remaining — setResonance is consumed at deployment
GBX	—	none — setMinter is single-use and permanently locks
Fund	none	none — contract is not Ownable
Strategy	none	none
BribeRouter	none	none
Bribe	none	addRewardToken , callable only by the immutable Resonance

SignalGBX , StrategyFactory , and BribeFactory retain a nominal Ownable owner after their one-time binding is consumed, but that owner has no remaining function to call. Resonance.setResonanceRouter is likewise single-use.

9.2 The four continuing administration actions

SELECTOR	TARGET	EFFECT	REVERSIBLE?
Resonance.addStrategy	Resonance	Deploys a Strategy, BribeRouter, and Bribe; registers payment token	No
Resonance.killStrategy	Resonance	Permanently excludes a Strategy from new signal and future revenue	No
Resonance.addBribeRewardToken	Resonance	Appends a reward token to a Strategy's Bribe, within the cap of 16	No
Resonance.setBribeBps	Resonance	Sets the global signaler share of acquisitions, bounded [0, 2000]	Yes

The first three are irreversible. addStrategy is reversible only in the sense that the created Strategy can later be killed; the deployed contracts persist forever. setBribeBps is the sole reversible action and the sole economic parameter: it is bounded in code and applies prospectively, so it cannot reclassify an amount already settled.

9.3 Authority explicitly absent

No contract in packages/contracts/src contains a proxy, Initializable , UUPSUpgradeable , Pausable , a delegatecall , a sweep or rescue function, a successor binding, a migration routine, an emission setter, a fee setter, a price oracle, an entropy source, or a claim-redirection path.

There is also no governance machinery: no Governor , no TimelockController , no generic executor, and no provider-specific adapter. ADR 0034 removed them, so the guarantees such a stack would have supplied — proposal filtering, quorum, voting periods, execution delay, cancellation rules — are supplied by nothing in this repository. They become properties of whichever external system is later given ownership of Resonance (§15.2).

10. Token and asset-flow graph

10.1 USDG

ORIGIN	PATH	TERMINAL STATE
Miner replacement payment	payer → Mine ; [p·8/10] retained as claim, remainder deposited → ResonanceRouter	Outgoing-tenure claim, or Router deposit
Revenue	ResonanceRouter → Resonance (on qualifying route() call) → Strategy	Auction inventory
Auction inventory	Strategy → buyer-nominated revenueReceiver	Exits the protocol
Rounding floors, zero-signal intervals, direct donations	remains in Resonance	Permanent surplus, unrecoverable
Direct donation to Router	remains in ResonanceRouter until a later qualifying route call	May wait indefinitely

10.2 GBX

ORIGIN	PATH	TERMINAL STATE
Initial supply	constructor creates no GBX	Zero
Mining issuance	Mine mints to slot occupant	Circulating
Signal	holder → SignalGBX custody, sGBX minted 1:1, committed to a Strategy	Escrowed, removable through SignalGBX
Direct donation to sGBX	remains in SignalGBX	Stranded surplus, no receipt
GBX-denominated auction	buyer → Strategy → Fund , burnable by anyone	Destroyed when burned
Redemption	redeemer → Fund → burned	Destroyed

10.3 Acquired assets and Bribe reward tokens

ORIGIN	PATH	TERMINAL STATE
Auction payment (Fund complement, 80–100%)	buyer → Strategy → Fund , atomically with the purchase	Fund backing until redeemed
Auction payment (Bribe share, 0–20%; 10% default)	buyer → Strategy → BribeRouter → paired Bribe	Buffered, then streamed to signalers
Compatible direct Router donation	donor → BribeRouter → paired Bribe on a qualifying call	Joins the next complete-buffer notification
Independent Bribe funding	funder → Bribe → signalers	Streamed reward; floors remain surplus
Direct donation to Bribe	donor → Bribe	Unscheduled surplus
Unsolicited transfer to Fund	anyone → Fund	Fund backing, unreviewed
Zero-supply Bribe emission	remains in Bribe	Unallocated surplus, unrecoverable

Structural invariant of the flow graph: the only paths *out* of Fund are redeem and burnGBX . There is no path from Fund to any administrator, and no caller-selectable destination for any share of an auction payment.

11. GBX

GBX is ERC20, ERC20Permit. Name "GumBall6900", symbol "GBX", 18 decimals. It deliberately does **not** inherit ERC20Votes : governance weight exists only as sGBX (§13).

11.1 State

VARIABLE	TYPE	MEANING
minter	address	Current mint authority
minterLocked	bool	Whether the one-time Mine handoff has completed permanently
lifetimeMinted	uint256	Cumulative raw units created by Mine
lifetimeBurned	uint256	Cumulative raw units destroyed

11.2 Initial supply

```
GBX.totalSupply() = GBX.lifetimeMinted() = GBX.lifetimeBurned() = 0
```

The constructor creates no tokens. It installs only a temporary setup minter that can perform the one-time Mine binding; mint remains disabled until that binding is permanently locked. There is no team allocation, presale, liquidity premit, vesting schedule, treasury reserve, or airdrop in the token contract.

11.3 Supply identity

Identity I-1. At every block:

```
GBX.totalSupply() = GBX.lifetimeMinted() - GBX.lifetimeBurned()
```

Proof sketch. lifetimeMinted is incremented by exactly amount immediately before every _mint(account, amount) in mint, and lifetimeBurned by exactly amount immediately before every _burn. Both counters are monotonically non-decreasing and written nowhere else. ■

Verified by testFuzz_SupplyEqualsLifetimeMintedMinusBurned and invariant_GBXSplyReconcilesWithBurns.

There is no supply cap. Under the emission schedule of §12.5, cumulative issuance grows without bound in time.

11.4 Mint authority handoff

setMinter(newMinter) succeeds only when all of the following hold, and then permanently sets minterLocked = true :

```
msg.sender == minter
minterLocked == false
newMinter != 0 ^ newMinter != minter ^ newMinter.code.length > 0
IMine(newMinter).gbx() == address(this)      -- reciprocal identity, try/catch guarded
```

mint requires both msg.sender == minter and minterLocked == true. Therefore the deployment-time coordinator can never mint, and after the handoff exactly one immutable address can mint forever. burn touches neither field, so burning cannot reopen authority.

Limitation. The reciprocal check proves the candidate *claims* the same GBX. It cannot distinguish a malicious lookalike returning the expected value. This is finding **M-03**, an open High release gate (\$41.3).

12. Mining and issuance

Mine is an ownerless ReentrancyGuard. It is the sole GBX issuer after the handoff.

12.1 Fixed Mine economics

CONSTANT	SYMBOL	VALUE	UNITS
PRICE_MULTIPLIER	m	2	integer multiplier
MIN_INITIAL_PRICE	P _{min}	10 ⁶	raw USDG
MAX_INITIAL_PRICE	P _{max}	2 ¹⁹² −1	raw USDG
INITIAL_TPS	u ₀	64 · 10 ¹⁸	raw GBX per second
HALVING_PERIOD	Δ	5,961,600	seconds
TAIL_TPS	u _∞	1 · 10 ¹⁸	raw GBX per second

Mine stores its supplied USDG and ResonanceRouter without calling the Router during construction. ADR 0045 instead requires pinned post-deployment reads proving `Mine.usdg() == USDG`, `Mine.resonanceRouter() == ResonanceRouter`, and `ResonanceRouter.usdg() == USDG` before the permanent GBX minter handoff or market exposure.

These values are bytecode constants rather than constructor arguments. The constructor accepts only GBX, USDG, and ResonanceRouter. Constants also include `BPS = 10_000`, `PREVIOUS_MINER_BPS = 8_000`, `PRICE_DECAY_PERIOD = 3600`, and `SLOT_COUNT = 16`.

12.2 Slot state

```
struct Slot {
    uint256 epochId;           // monotonically increasing fill counter, starts at 1
    uint256 initialPrice;      // starting USDG price of the current auction
    uint256 auctionStartedAt;  // timestamp the current auction began
    uint256 lastAccruedAt;     // start of this tenure's unsettled accrual
    uint256 tps;               // raw GBX per second, locked for this tenure
    address miner;             // zero when empty
}
```

An empty slot is `{epochId: 1, initialPrice: Pmin, auctionStartedAt: now, lastAccruedAt: now, tps: 0, miner: 0}`.

12.3 Replacement price

Formula F-1 (slot price). For elapsed $e = \text{now} - \text{auctionStartedAt}$ and $D = 3600$:

```
price(e) = initialPrice - [initialPrice · e / D]    for e < D
price(e) = 0                                       for e ≥ D
```

Symbols. `initialPrice`, raw USDG units, `uint256` bounded by $2^{192}-1$; `e`, `D` in seconds. *Rounding.* The subtracted term floors, so `price(e)` lies at or **above** the exact real-valued line — rounding favors the protocol and the outgoing tenure miner, never the incoming payer. `price` is a non-increasing step function. *Overflow.* `Math.mulDiv` at 512-bit intermediate precision; with `initialPrice` $< 2^{192}$ and `e` $< 2^{256}$, no overflow.

Worked example. `initialPrice = 2_000_000` (2.000000 USDG). At `e = 900`: `price = 2_000_000 - [2_000_000·900/3600] = 2_000_000 - 500_000 = 1_500_000`. At `e = 1800`, `1_000_000`; at `e = 2700`, `500_000`; at `e = 3600`, `0`. This is the curve reproduced in `packages/simulations/fixtures/economic-scenarios.json`.

12.4 Payment allocation

Formula F-2 (replacement split). For clearing price `p` and previous occupant `M`:

```
if p = 0:                claim = 0,                revenue = 0
if p > 0 and M = 0 (empty): claim = 0,                revenue = p
if p > 0 and M ≠ 0 (occupied): claim = [p · 8000 / 10000], revenue = p - claim
```

Rounding. `revenue = p - [0.8p] = [0.2p]`. The rounding unit accrues to the **protocol**, not the outgoing tenure miner. *Dust.* At most 1 raw USDG unit per replacement, always in the protocol's favor. There is no accepted loss.

OCCUPIED-SLOT REPLACEMENT



80% is a claim the outgoing tenure miner withdraws.
20% is deposited in Router for later routing.

EMPTY SLOT — NOBODY TO COMPENSATE



MINING REPLACEMENT. STARTING A NEW OCCUPIED-SLOT TENURE COMPENSATES THE OUTGOING MINER; TAKING AN EMPTY SLOT FUNDS THE PROTOCOL ENTIRELY.

Worked example. `p = 1_000_000`: `claim = 800_000`, `revenue = 200_000`. `p = 1_000_003`: `claim = [800_002.4] = 800_002`, `revenue = 200_001`. Note `200_001 > 0.2·1_000_003 = 200_000.6`, confirming the ceiling behavior.

Identity I-2 (Mine solvency).

```
USDG.balanceOf(Mine) ≥ Mine.totalClaimableMinerPayments()
```

with equality when no unsolicited donation has been made. `claimableMinerPayment[a]` and `totalClaimableMinerPayments` are incremented together in `_allocatePayment` and decremented together in `claimMinerPayment`; the Router deposit leaves Mine in the same transaction it arrives. Verified by `invariant_MineIsSolventAgainstReplacementClaims` and `testFuzz_MineRevenueAndHandoffClaimsReachFinalDestinationsWithoutDust`.

Mine/Router boundary (ADRs 0044 and 0049). `_collectAndDeposit` requests payer → Mine and Mine → ResonanceRouter transfers through `SafeERC20`, then emits `RevenueDeposited(slotIndex, epochId, amount)`. It does

not inspect balance deltas or call `route()`. The event records the nominal protocol share requested into the Router; standard canonical USDG semantics make that the amount delivered. `ResonanceRouter`'s distinct `RevenueRouted(caller, amount)` proves a later forward into Resonance. A failed USDG call into the Router still reverts the paid replacement, but a later Router or Resonance failure cannot. Permissionless routing has no caller role, bounty, or liveness guarantee, so the deposit may wait indefinitely. Optional manual, frontend, volunteer-keeper, cron, or future helper composition is periphery and cannot become a Mine correctness dependency.

12.5 Global emission schedule

Formula F-3 (global rate). Let $t_0 = \text{startTime}$, the immutable Mine-deployment timestamp, and let $\Delta = 69 \text{ days} = 5,961,600 \text{ seconds}$:

$$k(t) = \lfloor (t - t_0) / \Delta \rfloor$$

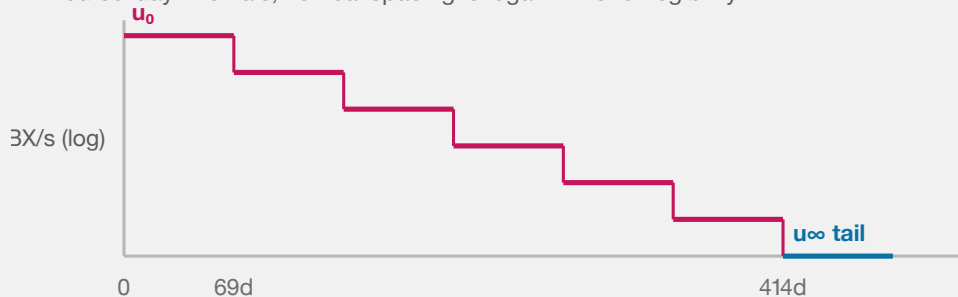
$$u(t) = \max(u_0 \gg k(t), u_\infty)$$

The global rate is prospective: it is read only when a slot begins a new tenure. A time boundary never reprices an occupied slot. Neither `totalMined` nor `pendingEmission()` selects the rate.

Symbols. $u(t)$ in raw GBX per second; t , t_0 , and Δ in seconds. **Arithmetic.** Solidity checked subtraction rejects a timestamp before `startTime`; ordinary block timestamps cannot produce that state. A right shift by a large count yields zero and the final `max` clamps the result to u_∞ , so the calculation is constant time and has no loop.

GLOBAL RATE AGAINST TIME SINCE MINE DEPLOYMENT

Fixed 69-day intervals; vertical spacing is logarithmic for legibility.



The provisional clock advances even while slots are empty; at day 414 the prospective rate reaches the tail.

THE PROSPECTIVE RATE HALVES EVERY 69 DAYS FROM MINE DEPLOYMENT. AT DAY 414 IT REACHES THE PERMANENT POSITIVE TAIL.

Fixed schedule. $u_0 = 64 \text{ GBX/second} = 230,400 \text{ GBX/hour}$. The first prospective halving occurs exactly 69 days after deployment, whether or not any slot has been occupied; later boundaries occur at the same fixed interval. The prospective path is 64, 32, 16, 8, 4, 2, then 1 GBX/second. The tail begins at the sixth boundary, day 414, when $64 / 2^6 = u_\infty = 1 \text{ GBX/second}$.

Synchronized reference only. If every slot is occupied from deployment, all sixteen slots are refreshed and settled exactly at every boundary, and no GBX is burned, the six pre-tail eras emit and supply 751,161,600 GBX by day 414. The scheduled tail flow is 31,536,000 GBX per 365-day year, initially about 4.1982% of that reference supply and declining as supply grows. This is not a cap, actual-supply forecast, or guaranteed inflation rate: legacy tenures can emit above that path, empty slots can emit below it, and burns change the live denominator.

BOUNDARY	ELAPSED DAYS	FRESH GLOBAL RATE	SYNCHRONIZED GROSS SUPPLY
Launch	0	64 GBX/s	0
1	69	32 GBX/s	381,542,400
2	138	16 GBX/s	572,313,600
3	207	8 GBX/s	667,699,200
4	276	4 GBX/s	715,392,000
5	345	2 GBX/s	739,238,400
6 (tail)	414	1 GBX/s	751,161,600

From the day-414 tail, the same no-burn reference reaches **782,697,600** GBX after one year, **814,233,600** after two years, **908,841,600** after five years, and **1,066,521,600** after ten years.

Accepted timing consequences. Time between deployment and public launch consumes the schedule. A replacement just before a boundary can lock the older rate for that tenure, while a replacement just after it receives the lower rate. These are direct consequences of a deployment-time clock combined with tenure locking.

Independent economic review remains open. ADRs 0042 and 0043 record the 64 GBX/second initial rate, 69-day period, and 1 GBX/second tail as provisional development values. They do not constitute audit approval, deployment authorization, or a signed production manifest. This is finding **M-04**.

12.6 Tenure-locked accrual

Formula F-4 (slot accrual). For an occupied slot,

```
pendingEmission(i) = (now - slots[i].lastAccruedAt) * slots[i].tps
pendingEmission() = storedPendingEmission + (now - pendingUpdatedAt) * aggregateTps
effectiveTotalSupply = GBX.totalSupply() + pendingEmission()
```

`aggregateTps` is the sum of the sixteen tenure rates. A replacement first advances the system accumulator at that old aggregate rate, then mints only the outgoing slot's accrued amount and subtracts the same amount from stored pending emission. Unrelated slots are neither iterated nor mutated.

Formula F-5 (new tenure rate). On a fill, after syncing pending emission and settling the outgoing slot:

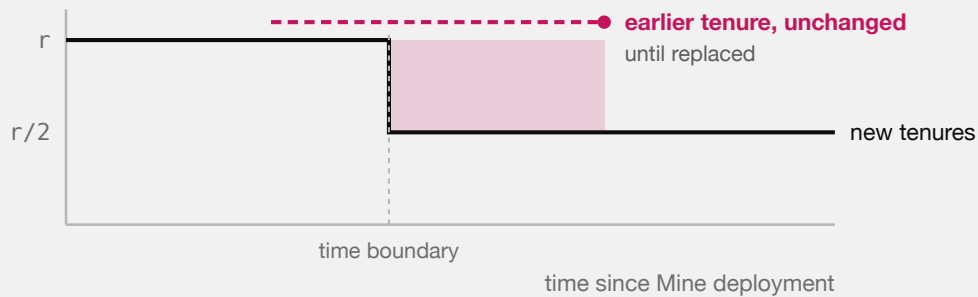
```
newSlot.tps = ⌊ globalTps(block.timestamp - startTime) / 16 ⌋
```

Rounding. The division residue is **unissued** — it is never minted to anyone. This is a deliberate, permanent reduction in aggregate issuance relative to the undivided global rate, bounded by 15 raw units per second across all slots.

Invariant. `slot.tps` is assigned in exactly one place in the contract: the `Slot` struct literal inside `mine()`. No other replacement, time boundary, or redemption modifies it. Verified by `test_TimeBasedHalvingNeverRepricesAnIncumbent`, `test_GlobalRateHalvesByDeploymentTimeEvenWhenEverySlotIsEmpty`, `testFuzz_CachedAccumulatorMatchesNaiveSlotsAndIndependentEconomicModel`, and `invariant_MiningPendingAndTpsCachesMatchEverySlot`.

Accepted consequence (finding M-01). Because outgoing tenures retain their assigned rate while new tenures divide the current global rate by sixteen, aggregate issuance can exceed the current global rate after a halving. The excess persists until the legacy tenures turn over.

A HALVING DOES NOT REPRICE ANYONE



The shaded band is real: while pre-halving tenures survive, total issuance sits above the current global rate. Accepted deliberately, so that nobody can change the deal a miner already paid for.

TIME-BASED HALVINGS APPLY TO NEWLY TAKEN SLOTS, NEVER TO SITTING MINERS. EXCESS ISSUANCE PERSISTS UNTIL THE LEGACY TENURE IS REPLACED, WHICH IS NOT GUARANTEED.

12.7 Next opening price

Formula F-6.

```
nextInitialPrice = clamp( [p · m / 1018] , P_min , 2192 - 1 )
```

A fill at $p = 0$ yields $[0] = 0$, which clamps up to $P_{\min}$. Recovery from the floor is therefore geometric at rate m per fill, not immediate. Verified by `test_AFreeFillAtFullDecayRestartsAtTheConfiguredFloor` and `test_RecoveryFromTheFloorIsOnlyGeometric`.

12.8 Fixed slot topology

Mine constructs exactly sixteen empty slots and exposes no owner or capacity mutation. This permanent topology maps directly to a 4-by-4 market. Pending-emission and redemption-supply reads remain constant time regardless of how many slots are occupied.

16 PERMANENT SLOTS

fixed at construction · no owner · no capacity setter

held 1.00×	held 1.00×	for sale price falling	held 0.50×
held 1.00×	held 0.50×	held 0.50×	held 0.25×
held 0.50×	empty deposits 100%	held 0.25×	held 0.25×
held 0.25×	held 0.25×	held 0.25×	held 0.25×

Each slot runs its own auction, falling to zero over 1 hour, and pays 80% to whoever it displaces. A slot's issuance rate is fixed when it is taken and never moves again. The mixed rates above are incumbents who took their slots under different halvings.

THE MINING MARKET IS A FIXED BOARD OF SIXTEEN INDEPENDENT SLOTS. NOTHING CAN ADD A SEVENTEENTH, AND NO SLOT'S RATE CAN BE CHANGED BY ANYONE ONCE IT IS TAKEN.

PENDING EMISSION, ALL SIXTEEN SLOTS

Different start times, different locked rates, all accruing at once.



$$\text{pendingEmission} = \text{stored} + (\text{now} - \text{updatedAt}) \times \text{aggregateTps}$$

The total is one multiplication no matter how many slots are occupied, which is why redemption can count unminted mining without touching the mine.

A replacement mints for the replaced slot only. The other fifteen are not read, checkpointed, or disturbed.

SIXTEEN INDEPENDENT TENURES, ONE CONSTANT-TIME TOTAL. REDEMPTION READS THE ACCUMULATOR; A REPLACEMENT SETTLES ONLY THE SLOT BEGINNING A NEW TENURE.

13. SignalGBX

SignalGBX is ERC20, ERC20Votes, ReentrancyGuard, Ownable. Name "SignalGumBall6900", symbol "sGBX", 18 decimals.

13.1 Non-transferability

```
function _update(address from, address to, uint256 value) internal override(ERC20, ERC20Votes) {
    if (from != address(0) && to != address(0)) revert TransferDisabled();
    super._update(from, to, value);
}
```

Only mint (`from == 0`) and burn (`to == 0`) are permitted. `transfer`, `transferFrom`, self-transfer, and zero-value transfer all revert, with or without an allowance.

13.2 Collateralization

Identity I-3.

$$\text{SignalGBX.totalSupply()} \leq \text{GBX.balanceOf(SignalGBX)}$$

with equality absent direct GBX donations. `_depositAndMint` requests `amount` GBX through `SafeERC20` and mints `amount` sGBX; `_burnAndWithdraw` burns `amount` and requests `amount` GBX out. ADR 0049 removes sender/receiver balance snapshots and trusts the canonical GBX implementation to move the requested amount. Direct GBX donations to the contract are stranded surplus that creates no receipt, signal, withdrawal entitlement, or voting power.

Verified by `testFuzz_SignalMoveWithdrawRoundTripIsLossless`, `invariant_SignalReceiptIsFullyCollateralized`, and `test_DirectDonationIsSurplusAndCreatesNoSignalVotesOrWithdrawalEntitlement`.

13.3 Mandatory signal-backing

ADR 0031 removed the separate `allocatedBalance` ledger, `_allocate`, `_deallocate`, and the `ISignalGBXAllocation` interface. Minting and burning are atomically coupled to the matching Resonance and paired-Bribe virtual-balance change, so **an idle receipt state is unreachable**.

Identity I-4 (mandatory signal-backing). Across live and killed Strategies, at every block:

$$\begin{aligned} \text{SignalGBX.balanceOf}(a) &= \sum_s \text{Bribe}(s).\text{signalWeightOf}(a) \\ \text{SignalGBX.totalSupply}() &= \sum_s \text{Bribe}(s).\text{totalSignalWeight}() \\ \text{GBX.balanceOf}(\text{SignalGBX}) &\geq \text{SignalGBX.totalSupply}() \end{aligned}$$

The account aggregate is read directly from `SignalGBX.balanceOf(a)`; Resonance exposes no duplicate aggregate or account-by-Strategy signal getter. There is no reachable successful state in which a newly minted raw unit is idle, or a burned raw unit leaves signal behind.

Verified by `invariant_EveryReceiptUnitIsAssigned`, `invariant_SignalWeightNeverExceedsTheReceiptBalance`, `invariant_AccountWeightsSumToAllRecordedStrategyWeight`, and `test_RemovedIdleReceiptSelectorsAreAbsentFromRuntime` — which asserts the removed selectors are absent from deployed **runtime**, not merely from source.

Scalar and batched additions/removals preserve both sides of I-4 atomically. A failed batch rolls back its complete aggregate custody and every earlier allocation.

13.4 Voting

`SignalGBX` inherits `ERC20Votes` on OpenZeppelin's default block-number clock (`CLOCK_MODE = mode=blocknumber`). Inside `_depositAndMint`:

```
if (delegates(account) == address(0)) _delegate(account, account);
```

so a first signal activates voting weight without a second transaction. An account with an existing non-zero delegate keeps it; an account that explicitly delegated to zero re-self-delegates on its next signal.

`SignalGBX` has **no ERC-2612 permit**. It inherits `EIP712` solely for ERC20Votes delegation signatures. This is deliberate: a permit authorizes spending, and `sGBX` cannot be spent.

Consequence of I-4 for a future integration. Because no `sGBX` can be idle, the ERC20Votes total supply is exactly the total signal committed across all Strategies. Any external system that uses `getPastTotalSupply` as a quorum denominator therefore measures against economically active weight only, with no idle receipts inflating it. That is a useful token property, not a quorum guarantee — the core defines no quorum (§15.2, §15.3).

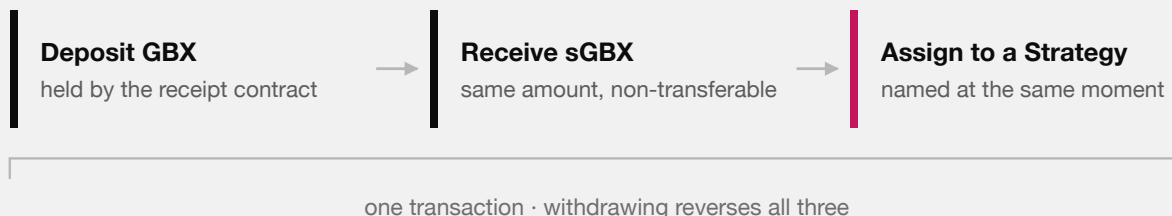
13.5 No lock

There is no timestamp, epoch, cooldown, or lock state anywhere in `SignalGBX`. Adding and removing signal may occur in consecutive blocks or the same transaction. Combined with checkpoints that survive removal (§15.3), this means an external system must not assume that recorded voting weight implies a currently held position. §35.3 and §38.2 analyze the consequence.

14. Signaling lifecycle

SIGNALING IS ONE STEP, NOT THREE

All of this happens in a single transaction, or none of it happens.



There is deliberately no state in between. You cannot hold `sGBX` that is not pointed at something, so every unit of signal that exists is a unit of signal doing work.

DEPOSIT, RECEIPT, AND COMMITMENT ARE ONE ATOMIC STEP. THE ABSENCE OF AN IN-BETWEEN STATE IS WHAT MAKES THE AMOUNT OF SIGNAL THAT EXISTS AND THE AMOUNT DOING WORK THE SAME QUANTITY.

14.1 Entry points

The complete user-facing surface is **four** functions on `SignalGBX`: bounded scalar add/remove operations and their optional struct-array variants. Every amount is an absolute raw-unit delta.

FUNCTION	COMPOSITION
<code>addSignal(strategy, amount)</code>	<code>_depositAndMint</code> → <code>Resonance.addSignalFor</code> → <code>Bribe.addSignalWeight</code>
<code>addSignalMany(allocations)</code>	checked sum → one deposit/mint → one canonical add hook and event per allocation
<code>removeSignal(strategy, amount)</code>	<code>Resonance.removeSignalFor</code> → <code>Bribe.removeSignalWeight</code> → <code>_burnAndWithdraw</code>
<code>removeSignalMany(allocations)</code>	checked sum → one canonical remove hook and event per allocation → one aggregate burn/GBX return

Every failed sub-operation reverts the complete transition.

An empty batch or any zero amount reverts. Duplicate Strategies execute sequentially. Batch length is caller-controlled, so interfaces simulate and split oversized requests; scalar removal remains the bounded exit. ADR 0051 removes `signal`, `signalWithPermit`, `moveSignal`, and `withdrawSignal` without compatibility aliases. SignalGBX consumes no permit signature. Smart wallets may atomically batch GBX approval with direct SignalGBX calls while remaining `msg.sender`; plain externally owned accounts establish allowance separately.

14.2 Sole-coordinator restriction

`Resonance.addSignalFor` and `removeSignalFor` carry `onlySignalGBX`, reverting `UnauthorizedSignalSource` for any other caller. Resonance exposes no dedicated move hook, and there is no shared write-through signal Router: such a Router would become the GBX, sGBX, and signal owner under `msg.sender` semantics. There is deliberately no second user-facing coordinator and no direct-signaling path on Resonance.

14.3 Canonical state ownership

Each level of the signal ledger has exactly one owner; no value is duplicated across contracts.

LEVEL	CANONICAL STORAGE AND PUBLIC ACCESSOR
Account aggregate	<code>SignalGBX.balanceOf(a)</code>
Account × Strategy	<code>Bribe(s).signalWeightOf(a)</code>
Strategy total	<code>Bribe(s).totalSignalWeight()</code>
Active total (live)	<code>Resonance.totalSignalWeight()</code>

The account aggregate is the sGBX balance itself, not a second ledger: ADR 0031 removed `allocatedBalance` precisely because it would always have been identical to `balanceOf` (§13.3).

14.4 Checkpoint-before-mutate ordering

This ordering is the protocol's defense against same-transaction revenue capture (finding A-11).

OPERATION	CHECKPOINT PERFORMED BEFORE ITS WEIGHT MUTATION
<code>addSignalFor</code>	<code>_updateRevenue(strategy)</code>
<code>removeSignalFor</code>	<code>_updateRevenue(strategy)</code>
<code>killStrategy</code>	<code>_updateRevenue(strategy)</code> before validation and mutation
<code>notifyRevenue</code>	<code>_updateRevenue(address(0))</code> — global index only
<code>distributeRevenue</code>	<code>_updateRevenue(strategy)</code>
<code>Strategy.buy</code>	<code>Resonance.distributeRevenue(address(this))</code> before inventory

Additionally, `Bribe.addSignalWeight` and `Bribe.removeSignalWeight` each call `_updateAllRewards(account)` before mutating `totalSignalWeight` or `signalWeightOf`. There is no boundary carry or Fund-classification pass.

Consequence. In a single transaction, no stream time elapses between operations, so the revenue/reward indices cannot advance. A flash-signal therefore accrues exactly zero newly-notified revenue and zero newly-notified Bribe rewards. A signal held across real elapsed time legitimately earns that interval's flow — the protocol provides no epoch, cooldown, or anti-churn guarantee beyond this ordering.

Verified by `test_FlashSignalWeightCannotRedirectANewNotification`, `test_FlashSignalWeightCannotStealAccruedBribeRewards`, `test_NewStrategyWeightReceivesOnlyPostEntryRevenue`, and `test_StrategyAddedAfterAc-`

`cruaCannotClaimHistoricRevenue` (finding A-11).

15. Protocol administration and external governance

ADR 0034 removed `ProtocolGovernor` and the protocol `TimelockController` from the core. This section specifies the administration surface that remains and states precisely which guarantees the core does **not** provide as a result.

15.1 The complete owner-gated surface

`Resonance` is the only core contract with continuing custom owner authority. `SignalGBX`, `StrategyFactory`, and `BribeFactory` retain setup-only Ownable shells until production explicitly renounces them after their one-time `Resonance` bindings are consumed; those shells expose no custom protocol action after setup. `Resonance`'s owner-gated functions are:

FUNCTION	SOURCE	EFFECT	REVERSIBLE?
<code>addStrategy(IERC20, Config)</code>	<code>Resonance.sol</code>	Deploys <code>Strategy</code> , <code>BribeRouter</code> , and <code>Bribe</code> ; registers the payment token	No (kill only)
<code>killStrategy(address)</code>	<code>Resonance.sol</code>	Permanently excludes a <code>Strategy</code> from new signal and future revenue	No
<code>addBribeRewardToken(address, address)</code>	<code>Resonance.sol</code>	Appends a reward token to a <code>Strategy</code> 's <code>Bribe</code> , within <code>MAX_REWARD_TOKENS</code>	No
<code>setBribeBps(uint256)</code>	<code>Resonance.sol</code>	Sets the global signaler share, bounded <code>[0, MAX_BRIBE_BPS]</code> (ADR 0036)	Yes, prospectively
<code>setResonanceRouter(address)</code>	<code>Resonance.sol</code>	Binds the sole <code>ResonanceRouter</code>	Single-use, then closed
<code>transferOwnership / renounceOwnership</code>	<code>OpenZeppelin Ownable</code>	Moves or destroys the owner role	Not by the protocol

`setResonanceRouter` reverts with `ResonanceRouterAlreadySet` after its first success, so it is a deployment binding rather than a continuing authority. The four continuing capabilities are therefore exactly `addStrategy`, `killStrategy`, `addBribeRewardToken`, and `setBribeBps`.

Enforced constraints on those calls. These are Solidity checks, not procedural expectations:

CONSTRAINT	MECHANISM
The final live <code>Strategy</code> cannot be killed	<code>if (liveStrategyCount == 1) revert FinalLiveStrategy(strategy)</code>
The signaler share can never exceed 20%	<code>if (newBribeBps > MAX_BRIBE_BPS) revert BribeBpsAboveMaximum()</code>
A rate change cannot reclassify a settled amount	Snapshotted per payment, before any payment-token interaction
A <code>Strategy</code> cannot be killed twice	<code>isStrategyLive</code> check, <code>StrategyAlreadyDead</code>
sGBX cannot be a payment token or <code>Bribe</code> reward token	<code>ForbiddenPaymentToken</code> , <code>ForbiddenRewardToken</code>
Payment and reward tokens must be deployed code	<code>code.length == 0</code> rejection
<code>Bribe</code> reward tokens are append-only and capped at sixteen	<code>Bribe.MAX_REWARD_TOKENS</code> , <code>RewardAlreadyAdded</code>
<code>Strategy</code> auction parameters are bounded at construction	<code>Strategy</code> constructor range checks (§21.1)

15.2 Authority the core does not implement

The core contains **no** Governor, Timelock, generic executor, multicall relay, or provider-specific governance adapter. It therefore makes none of the following guarantees, and no repository document should assert them:

ABSENT GUARANTEE	CONSEQUENCE
Selector-bounded proposal filtering	The owner calls the four continuing functions directly; no calldata filter exists
Proposal threshold, quorum, or support	The core defines none; <code>liveStrategyCount</code> is the only counting rule
Voting period or voting delay	The core defines none
Post-approval execution delay	Owner calls take effect in the calling transaction
Permissionless execution after a delay	Not applicable; there is no queue
Cancellation, veto, or guardian	No such role exists in the core
Sole-proposer closure	Not applicable
Immutable governance parameters	Not provided; <code>bribeBps</code> is mutable within its hard 0–20% bound

The owner address itself is the entire authority model in this development tree. Nothing in `packages/contracts/src` constrains who or what that address is.

15.3 SignalGBX voting checkpoints

SignalGBX is ERC20, ERC20Votes, ReentrancyGuard, Ownable. It retains vote checkpoints deliberately, for a future external integration, and the core assigns them no semantics.

PROPERTY	VALUE
Interface	IVotes via OpenZeppelin ERC20Votes
Clock	Default block-number clock; no <code>clock()</code> or <code>CLOCK_MODE</code> override
Transferability	None — <code>_update</code> reverts <code>TransferDisabled</code> unless <code>from</code> or <code>to</code> is zero
Delegation	Standard; <code>_depositAndMint</code> self-delegates any account with no delegate
Approval permit	Not implemented on sGBX; the underlying GBX carries <code>ERC20Permit</code>
Supply relationship	Every sGBX unit is backed one-for-one by escrowed GBX and assigned to one Strategy (\$14)

Because `_depositAndMint` self-delegates on first deposit, an account that has never delegated still accrues voting weight from its first signal without a second transaction. This removes the former undelegated-supply liveness concern **as a property of the token**; it does not constitute a quorum guarantee, because the core defines no quorum.

Checkpoints survive removal. `removeSignal` and `removeSignalMany` burn sGBX and write a new checkpoint, but historical checkpoints at earlier blocks are immutable by construction. An account may acquire GBX, signal it, allow a block to pass, remove signal, and retain its recorded weight at that past block. Whether that is exploitable depends entirely on whether the selected external system reads historical balances and how it spaces its snapshot from its voting window. This is finding **G-01**, retained as an integration property rather than a core defect.

15.4 Requirements on the future integration

ADR 0034 makes deployment conditional on a later ADR that pins and reviews at least:

1. provider, exact release, deployed bytecode, and proxy or upgrade model;
2. plugin set, permission graph, root and admin holders, and any emergency path;
3. direct compatibility with SignalGBX voting checkpoints and delegation, including snapshot-to-vote spacing (G-01);
4. proposal creation, quorum, support, voting duration, execution, batching, cancellation, and delay semantics; and
5. the exact `Resonance` owner address and the transaction evidence proving the handoff.

Until every item is selected, tested, and recorded, no deployment is authorized. Aragon is under consideration; no provider is part of the reviewed protocol graph, and this document must not be read as selecting one.

15.5 Security consequences

- Analysis of snapshot borrowing, quorum liveness, permission graphs, upgrade authority, and execution behavior does not apply to this repository and must be repeated against the selected system’s exact release.
- A compromised or careless `Resonance` owner can add Strategies, kill any Strategy except the last live one, register reward tokens up to the sixteen-token cap, transfer ownership, or renounce it. `renounceOwnership` would permanently freeze the Strategy set at its current membership.
- Owner authority does not reach mining parameters, mint authority, Fund assets, liquidity custody, auction mechanics, or the sixteen-slot count. Those are immutable or held by ownerless contracts (§9). The one economic parameter it does reach, the signaler share, is bounded in code to `[0, MAX_BRIBE_BPS]` and applies prospectively only, so no owner can reclassify a settled amount or drive Fund’s cumulative share below 80%.
- A production deployment that retains the temporary setup owner is a protocol with an ordinary admin key. Removing that owner is a release gate, not a recommendation (findings **M-03**, **G-01**, **G-03**).

16. Resonance

`Resonance` is `ReentrancyGuard`, `Ownable`. It is a Synthetix-shaped rewarder in which the “stakers” are Strategies and the “stake” is signal weight.

16.1 State

```
struct RevenueData {
    uint256 periodFinish;           // exclusive end of the active seven-day schedule
    uint256 revenueRate;           // whole raw USDG units emitted per second
    uint256 lastUpdateTime;        // last timestamp folded into revenuePerSignalStored
    uint256 revenuePerSignalStored; // cumulative index, 1e36 scaled
}
```

`Resonance` specializes the `Bribe` lineage to one immutable reward token: USDG. It stores one scalar `revenueData` schedule plus `strategyRevenuePerSignalPaid[strategy]` and `strategyRevenue[strategy]`; there is no `Resonance` reward-token registry, token-keyed schedule, or second-token extension point. `Bribe` remains independently multi-token.

Constants: `REWARD_DURATION = 7 days = 604_800`, `REWARD_PRECISION = 1036`.

16.2 Live-weight semantics

`totalSignalWeight` is the sum of `Bribe(s).totalSignalWeight()` over **live** Strategies only. It is:

- incremented by `amount` in `addSignalFor`;
- decremented by `amount` in `removeSignalFor` **only if the Strategy is alive**;
- decremented by the Strategy’s entire recorded weight in `killStrategy`.

This asymmetry is precisely what prevents a killed Strategy’s weight from being subtracted twice.

17. Intended six-decimal USDG revenue mathematics

This section is the protocol's most precision-sensitive arithmetic. Under the intended deployment, USDG carries six decimals while the weight denominator carries eighteen; contracts operate only on raw units and do not read or enforce USDG decimals.

17.1 The decimal problem

Under the intended deployment, USDG has 6 decimals and sGBX has 18. A naive Synthetix index at $1e18$ precision would compute, for E raw USDG emitted against total weight W :

$$\Delta_{\text{index}} = \lfloor E \cdot 10^{18} / W \rfloor$$

With $W = 4600 \cdot 10^{18}$ (a modest 4,600 sGBX) and $E = 10^6$ raw units (1.00 USDG), this yields $\lfloor 10^6 \cdot 10^{18} / 4.6 \cdot 10^{21} \rfloor = \lfloor 217.4 \rfloor = 217$, and the per-Strategy recovery $\lfloor \text{weight} \cdot 217 / 10^{18} \rfloor$ reintroduces a second floor. The compounding of two floors at insufficient precision would round small allocations entirely to zero.

Resolution. $\text{REWARD_PRECISION} = 10^{36}$, chosen so the index carries $10^{36} / 10^{18} = 10^{18}$ units of precision per unit of weight — restoring 18 significant digits of headroom against a 6-decimal reward.

17.2 Index formulas

Formula F-9 (cumulative index).

```
revenuePerSignal() = revenuePerSignalStored           if W = 0
                   = revenuePerSignalStored + mulDiv(E, 1036, W) otherwise

where W = totalSignalWeight
      E = emissionBetween(lastUpdateTime, min(now, periodFinish))
```

Symbols. E in raw USDG units; W in raw sGBX units (18 decimals); index in $1e36$ -scaled USDG-per-raw-sGBX. *Rounding.* Floor. The residue $E \cdot 10^{36} \bmod W$ is **discarded, not carried** (§17.5). *Overflow.* `mulDiv` computes the 512-bit product $E \cdot 10^{36}$ before dividing. E is bounded by the scheduled amount; even $E = 2^{96}$ leaves the product below 2^{216} . No overflow is reachable.

Formula F-10 (Strategy entitlement).

```
earnedRevenue(s) = strategyRevenue[s]
                  + mulDiv(activeWeight(s), revenuePerSignal() - strategyRevenuePerSignalPaid[s], 1036)

where activeWeight(s) = isStrategyLive[s] ? Bribe(bribeFor[s]).totalSignalWeight() : 0
```

Rounding. Floor. The residue $\text{activeBalance} \cdot \Delta \bmod 10^{36}$ is **discarded, not carried** (§17.5). *Note.* A killed Strategy has $\text{activeWeight} = 0$, so `earnedRevenue` returns exactly its stored pre-kill amount forever.

17.3 The raw emission schedule

Formula F-11 (Synthetix-shaped schedule restart). On a qualifying notification of `amount` at time t_0 , let the old schedule's whole-unit remainder be $\text{remaining} = \max(\text{periodFinish} - t_0, 0) \cdot \text{oldRevenueRate}$:

```

S          = amount + remaining
revenueRate = ⌊S / 604800⌋
periodFinish = t0 + 604800
lastUpdateTime = t0

```

Formula F-12 (emission between two timestamps). For applicable timestamps `from < to ≤ periodFinish` :

```
emissionBetween(from, to) = (to - from) · revenueRate
```

The schedule represents only whole per-second units:

```

scheduled      = 604800 · ⌊S/604800⌋
rateFloorSurplus = S - scheduled = S mod 604800

```

The remainder is neither front-loaded nor carried. It stays in Resonance as unallocated USDG surplus. The sole Router requires at least `REWARD_DURATION` raw units before notification, so a newly started schedule has `revenueRate ≥ 1` ; the Resonance contract itself remains a scalar Synthetix engine rather than maintaining a second remainder boundary.

Formula F-13 (remaining).

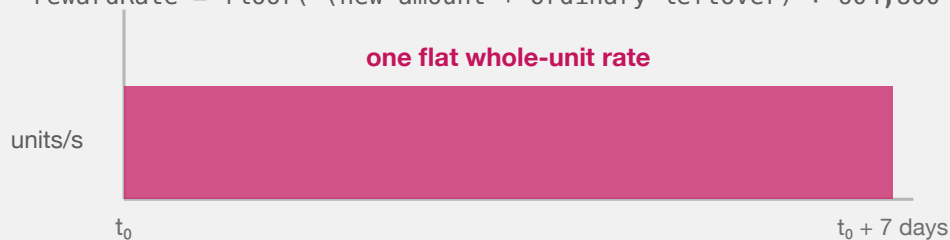
```

remainingRevenue() = 0                                if now ≥ periodFinish
                   = (periodFinish - now) · revenueRate otherwise

```

ONE SYNTHETIX-SHAPED SEVEN-DAY SCHEDULE

```
rewardRate = floor( (new amount + ordinary leftover) ÷ 604,800 )
```



$S \bmod 604,800$ is not front-loaded or carried; it remains unallocated contract surplus.

A qualifying active top-up rolls in left = remaining seconds × old rate, then restarts at now.

Routers wait for balance $\geq 604,800$ raw units and $\geq$ left before notifying.

RESONANCE AND BRIBE USE ONE FLAT WHOLE-UNIT RATE WITH ORDINARY LEFTOVER ROLLOVER; THE DIVISION REMAINDER STAYS AS UNALLOCATED SURPLUS.

Worked example A (clean division). $S = 604_800_000_000$ raw USDG (604,800.000000 USDG). `revenueRate = 1\_000\_000` . Emission is exactly 1 USDG per second for seven days, with zero rate-floor surplus.

Worked example B (with remainder). $S = 1_000_000$ raw (1.00 USDG). `revenueRate = ⌊1\_000\_000/604\_800⌋ = 1` ; the schedule emits `604\_800` raw units over seven days. The remaining `395\_200` raw units stay in Resonance as unallocated rate-floor surplus. Verified by `test_NotificationStartsOneScalarScheduleAndKeepsTheRateFloorAsSurplus` ,

`test_OrdinaryRateFloorLeavesTheRawRemainderAsSurplus` , and `test_RouterBuffersUntilAtLeastOneRawUnitPerSecondCanBeScheduled` .

17.4 Allocation worked example

Three Strategies, weights $w_A = 1000 \cdot 10^{18}$, $w_B = 3000 \cdot 10^{18}$, $w_C = 600 \cdot 10^{18}$, so $W = 4600 \cdot 10^{18}$. One day elapses under Worked example A's schedule, emitting $E = 86_400 \cdot 10^6 = 86_400_000_000$ raw USDG.

Index advance: $\Delta = \text{mulDiv}(86_400_000_000, 10^{36}, 4600 \cdot 10^{18}) = \lfloor 8.64 \cdot 10^{46} / 4.6 \cdot 10^{21} \rfloor = 18_782_608_695_652_173_913_043_478$ (units of $1e36$ -scaled USDG per raw sGBX).

Per-Strategy recovery:

STRATEGY	WEIGHT (RAW SGBX)	MULDIV(W, Δ, 10 ³⁶)	USDG
A	$1000 \cdot 10^{18}$	18_782_608_695	18,782.608695
B	$3000 \cdot 10^{18}$	56_347_826_086	56,347.826086
C	$600 \cdot 10^{18}$	11_269_565_217	11,269.565217
Sum		86_399_999_998	86,399.999998

Residue: 2 raw units (0.000002 USDG) unallocated. This is the compounded global-index and per-Strategy floor. It remains in Resonance permanently as surplus.

17.5 Accepted surplus — what Resonance deliberately does *not* conserve

Resonance has **four** arithmetic or time sources of permanently unallocated USDG, and carries none of them:

- Schedule-rate floor.** $(\text{amount} + \text{remaining}) \bmod \text{REWARD_DURATION}$ on each schedule restart (§17.3, F-11).
- Global-index floor.** $E \cdot 10^{36} \bmod W$ per checkpoint (§17.2, F-9).
- Per-Strategy floor.** $\text{activeBalance} \cdot \Delta \bmod 10^{36}$ per Strategy per checkpoint (§17.2, F-10).
- Zero-active-weight emission.** `revenuePerSignal()` returns early when $W = 0$, but `_updateRevenue` still advances `lastUpdateTime` to `_lastApplicableRevenueTime()` . Stream time therefore elapses with **no** Strategy credited, and that interval's emission is permanently unclaimable.

Plus a fifth category that is never scheduled at all: **direct USDG transfers to Resonance**, since scheduling occurs only inside `notifyRevenue` .

ADR 0047 applies the same simple rule to Resonance and Bribe: whole-unit floors remain surplus instead of introducing remainder schedules, carry buckets, Fund classification, or boundary-specific state. $1e36$ makes index floors small for intended magnitudes, but does not make them zero.

No exact conservation identity and no lifetime dust bound is claimed for Resonance. There is no synchronization, sweep, rescue, or later-allocation path. Checkpoint frequency and protocol lifetime determine the accumulation, and neither is bounded.

The strongest provable statement is an inequality (§30, Identity I-7).

18. Reward-period notification behavior

18.1 Router retention rule

`ResonanceRouter.route()` is permissionless and behaves as:

```

pending = USDG.balanceOf(router)
if pending == 0:            revert NoRevenue
minimum = max(Resonance.REWARD_DURATION(), Resonance.remainingRevenue())
if pending < minimum:      emit RevenueHeld(caller, pending, minimum); return 0
approve and notify the ENTIRE pending balance

```

The absolute `REWARD_DURATION`-raw-unit floor ensures the restarted whole-unit rate is nonzero. `remainingRevenue()` decays monotonically, but a balance below `REWARD_DURATION` never qualifies without further deposits. Qualification does not execute a transaction: someone must still call `route()`, and no role or bounty guarantees that call. A sub-threshold attempt returns without reverting; Mine does not attempt routing at all. The `remainingRevenue` gate also prevents cheap stream-reset griefing.

18.2 Resonance acceptance rule

`notifyRevenue(amount)` is `onlyResonanceRouter`, runs `_updateRevenue(address(0))` first, then:

```

if amount == 0:            revert ZeroAmount
remaining = remainingRevenue()
if amount < remaining:     revert RevenueBelowRemaining
pull `amount` from the router with SafeERC20
revenueRate = [(amount + remaining) / REWARD_DURATION]
restart the seven-day schedule at now          (F-11)

```

Only the bound Router can call this function, and the Router separately enforces `amount ≥ REWARD_DURATION`. Neither side duplicates sender and receiver balance snapshots; USDG is assumed to have standard non-rebasing ERC-20 semantics.

18.3 Economic consequence of the restart rule

A restart moves `periodFinish` to `now + 7 days` and sets the rate to `[(amount + remaining)/604800]`. This can **raise or lower** the instantaneous rate:

- If `amount` is large relative to `remaining`, the rate rises.
- If `amount ≈ remaining` and substantial time has elapsed, the rate can fall, because the same remaining value is re-spread over a fresh seven days.

An actor wishing to force an early reset through the Router must supply enough for its complete balance to reach both `REWARD_DURATION` and `remainingRevenue`, so the manipulation is economically self-financing rather than free. The residual timing influence is intentional and accepted (§34.3).

Verified by `test_QualifyingTopUpCheckpointsAndRestartsWithRewardPlusLeft`, `test_RouterBuffersUntilItsBalanceReachesTheActiveAmountLeft`, and `test_SubThresholdRevenueWaitsUntilTheRouterBalanceQualifies`.

18.4 Relationship to Bribe

ADR 0047 intentionally makes the two reward engines share the same upstream-shaped core:

BEHAVIOR ON FUNDING DURING AN ACTIVE STREAM	RESONANCE	BRIBE
Minimum new amount	Router enforces $\geq \text{REWARD_DURATION}$ and $\geq \text{remainingRevenue}$	notification enforces both directly
Effect on active schedule	checkpoint, add ordinary leftover, restart at <code>now</code>	same
Schedule-rate remainder	unallocated token surplus	unallocated token surplus
Zero-supply behavior	time elapses; interval is unallocated	same
Asset and cardinality	one immutable USDG schedule	up to sixteen registered token schedules

19. Signal checkpoint ordering

The ordering discipline of §14.4 has a precise formal statement.

Property P-1 (no same-transaction capture). Let τ be a transaction containing a weight mutation at internal step j and any revenue-bearing operation at step $k > j$. Because every mutation and every notification calls `_updateRevenue` before mutating, and because `revenuePerSignal` advances only as a function of `_lastApplicableRevenueTime()` – `lastUpdateTime`, which is identically zero within one `block.timestamp`, the index cannot advance between steps j and k . Therefore weight introduced at step j earns exactly zero from any emission recognized at step k .

Property P-2 (no retroactive redirection). Symmetrically, weight *removed* at step j retains its full entitlement to all emission checkpointed before step j , because `_updateRevenue(strategy)` settles `strategyRevenue[strategy]` into storage before the weight changes.

Corollary. The pair (P-1, P-2) means signal weight earns exactly the emission that elapses while it is allocated — no more, no less, up to the floors of §17.5.

What P-1 does not provide. It bounds capture within a transaction, not across blocks. A signaler who allocates and holds across real elapsed time earns that interval's flow legitimately, and may unallocate immediately afterwards. There is deliberately no epoch, cooldown, minimum duration, or anti-churn mechanism. Rapid allocation movement and wallet-splitting are permitted by design.

19.1 Strategy purchase ordering

`Strategy.buy` calls `IResonance(resonance).distributeRevenue(address(this))` before reading `usdg.balanceOf(address(this))`. Consequently:

- the buyer receives every USDG unit released to the Strategy through the execution timestamp, not merely its stale visible balance; and
- combined with P-1, a same-transaction signal-then-buy sequence can acquire only inventory that predated the routed payment.

This is the direct remediation of finding A-11.

20. Strategy registration and lifecycle

20.1 Registration

`Resonance.addStrategy(paymentToken, config)` is `onlyOwner nonReentrant` and performs, in order:

- Reject `paymentToken` that is zero or code-less; reject `paymentToken == signalGBX (ForbiddenPaymentToken)`.

2. `bribeFactory.createBribe()` → new `Bribe` bound to this `Resonance`; `Bribe` has no `Fund` dependency.
3. `bribe.addRewardToken(paymentToken)` — the payment token occupies reward slot 1 of 16 automatically.
4. `strategyFactory.createStrategy(usdg, paymentToken, fund, bribe, config)` → deploys `Strategy` and `BribeRouter` together.
5. Record `isStrategyRegistered`, `isStrategyLive`, `bribeFor`, and `bribeRouterFor`; increment `liveStrategyCount`.
6. Set `strategyRevenuePerSignalPaid[strategy] = revenueData.revenuePerSignalStored`.

Step 6 is what prevents a newly registered `Strategy` from claiming historical revenue: it starts at the current index, so its first `earnedRevenue` computation sees $\Delta = 0$. Verified by `test_StrategyAddedAfterAccrualCannotClaimHistoricRevenue`.

The rejection of `sGBX` as a payment token (finding **E-03**) exists because `sGBX` transfers are permanently disabled: a `Strategy` priced in `sGBX` could never be filled, and the reward slot it consumed in the append-only `Bribe` registry could never be reclaimed.

Both factories reject any caller other than their permanently bound `Resonance` (`NotResonance`), so there is no public `Strategy` or `Bribe` creation path.

20.2 Death

`killStrategy(strategy)` is `onlyOwner nonReentrant` and calls `_updateRevenue(strategy)` before validation:

```
require isStrategyRegistered[strategy] ^ isStrategyLive[strategy]
require liveStrategyCount != 1                -- else revert FinalLiveStrategy
isStrategyLive[strategy] ← false
--liveStrategyCount
totalSignalWeight ← totalSignalWeight - Bribe(bribeFor[strategy]).totalSignalWeight()
```

The `_updateRevenue` call runs first, so the `Strategy`'s accrued whole `USDG` units are settled into `strategyRevenue` and remain permanently claimable via `distributeRevenue`. Thereafter `earnedRevenue` uses an active weight of zero, so no further accrual occurs.

Death is **irreversible**. There is no revive function. Since ADR 0031, the **final** live `Strategy` cannot be killed at all: `liveStrategyCount == 1` reverts `FinalLiveStrategy`, so at least one valid signal destination always exists (§29.3).

20.3 Post-death signal semantics

OPERATION ON A KILLED STRATEGY <code>S</code>	PERMITTED?	EFFECT ON <code>TOTALSIGNALWEIGHT</code>
<code>addSignalFor(a, s, x)</code>	No	reverts <code>StrategyAlreadyDead</code>
<code>removeSignalFor(a, s, x)</code>	Yes	unchanged — weight was already excluded at kill

The asymmetry is essential: subtracting on `removeSignalFor` would double-subtract weight already removed by `killStrategy`. A wallet may later add the returned `GBX` to a live `Strategy` in the same account-level batch, but that is composition of two direct operations rather than a core move selector. Verified by `test_DeadStrategySignalCanExitWithoutSubtractingActiveSupplyTwice`, `test_MoveFromKilledStrategyReentersLiveWeightExactlyOnce`, and `test_MoveSignalDestinationFailureRollsBackSourceRemoval`.

20.4 The closed-pool consequence after ADR 0047

`Bribe` has no kill state and no awareness that its Strategy died. It becomes a **closed pool**:

- `addSignalWeight` is unreachable, because the only caller is `Resonance.addSignalFor`, which rejects dead Strategies.
- Incumbent signalers may remain, earn elapsed rewards, claim, and exit incrementally.
- Reward time never pauses. If supply becomes zero, any remaining scheduled emission elapses without an account being credited and stays as unallocated Bribe surplus.
- `notifyReward` remains callable when its duration and remaining-reward gates are met, even if no signaler can return.

The old BR-1 queue-and-pause terminal state was superseded by ADR 0047; there is no queue or paused state now. The economic warning remains: a notification to a dead, zero-supply Bribe can become entirely unclaimable, with no refund, rescue, sweep, or Fund reclassification. Interfaces must identify dead Strategies and must not imply recovery.

21. Reverse Dutch acquisition auctions

21.1 Immutable configuration

PARAMETER	SYMBOL	BOUND
<code>epochDuration</code>	<code>D_s</code>	$3600 \leq D_s \leq 31\,536\,000$ (1 hour ... 365 days)
<code>priceMultiplier</code>	<code>m_s</code>	$1.1 \cdot 10^{18} \leq m_s \leq 3 \cdot 10^{18}$
<code>minimumPrice</code>	<code>p_min</code>	$10^6 \leq p_{\min} \leq 2^{192-1}$
<code>initialPrice</code>	<code>p_0</code>	$p_{\min} \leq p_0 \leq 2^{192-1}$

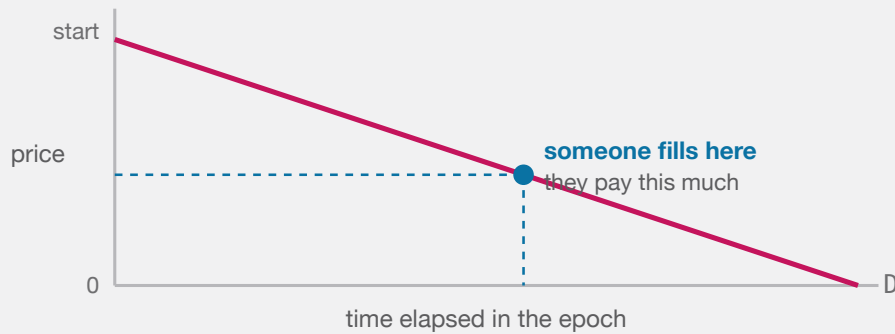
`PRICE_SCALE = 1018` , `ABSOLUTE_MINIMUM_PRICE = 106` , `ABSOLUTE_MAXIMUM_PRICE = 2192-1` .

21.2 Price function

Formula F-14. For `e = now - epochStartedAt` :

$$\begin{aligned} \text{currentPrice}(e) &= \text{initialPrice} - \lfloor \text{initialPrice} \cdot e / D_s \rfloor && \text{for } e < D_s \\ &= 0 && \text{for } e \geq D_s \end{aligned}$$

Units. Raw units of the **payment token**, whose decimals are that token's own. *Rounding.* Floor on the subtracted term, so the quoted price is at or above the ideal line. Monotonically non-increasing within an epoch. *Overflow.* `mulDiv` ; `initialPrice < 2192` bounds the intermediate.



The next epoch restarts at the price just paid $\times$ a fixed multiplier, so a hot auction opens higher and an unfilled one keeps falling to zero.

BOTH THE MINING SLOTS AND THE STRATEGY AUCTIONS USE THE SAME SHAPE: THE ASKING PRICE FALLS IN A STRAIGHT LINE TO ZERO, AND WHOEVER THINKS IT IS CHEAP ENOUGH FILLS IT.

Important distinction. $p_{\min}$ is a floor on the **next epoch's starting price**, not on the fill price. Fills below $p_{\min}$, including at exactly zero after full decay, are normal and expected.

Worked example. $D_s = 86_400$ (24 h), $p_0 = 4 \cdot 10^8$ raw (4.00000000 WBTC at 8 decimals). At $e = 47_520$ (55% elapsed): $\text{price} = 4 \cdot 10^8 - [4 \cdot 10^8 \cdot 47_520 / 86_400] = 4 \cdot 10^8 - 2.2 \cdot 10^8 = 1.8 \cdot 10^8 = 1.8$ WBTC.

21.3 Fill

```
buy(revenueReceiver, expectedEpochId, deadline, maximumPayment):
    require revenueReceiver != 0
    require now <= deadline                else DeadlinePassed
    require expectedEpochId == epochId    else EpochIdMismatch
    appliedBribeBps ← Resonance.bribeBps() -- before any token interaction
    Resonance.distributeRevenue(this)      -- pull released USDG first
    revenueAmount ← usdg.balanceOf(this)
    require revenueAmount != 0             else EmptyRevenue
    paymentAmount ← currentPrice()
    require paymentAmount <= maximumPayment else MaximumPaymentExceeded
    if paymentAmount != 0:
        pull paymentAmount with SafeERC20
        bribeAmount ← [paymentAmount · appliedBribeBps / BPS]
        transfer paymentAmount - bribeAmount directly to Fund
        if bribeAmount != 0: transfer it to BribeRouter
    transfer revenueAmount to revenueReceiver with SafeERC20
    initialPrice ← nextInitialPrice(paymentAmount)
    epochStartedAt ← now
    epochId ← epochId + 1
```

Buyer protections are `expectedEpochId` (defeats a competing fill changing the price mid-flight), `deadline`, and `maximumPayment`.

Formula F-15 (next starting price).

```
nextInitialPrice = clamp( [ paymentAmount · m_s / 1018 ], p_min , 2192 - 1 )
```

A zero-price fill yields `p_min`. Recovery is geometric at `m_s` per fill. Verified by `test_OneLateFillCollapsesTheAuctionToItsFloor` and `test_RecoveryFromTheFloorIsOnlyGeometric`.

21.4 Receiver semantics

`revenueReceiver` is buyer-chosen and unvalidated beyond non-zero. Consequences, all tested:

RECEIVER	OUTCOME
Ordinary address	Normal
Fund	USDG becomes Fund backing
Resonance	Creates unscheduled surplus — the USDG is not re-notified
The Strategy itself	A standard ERC-20 self-transfer succeeds; the USDG remains inventory for a later purchase

22. Auction-payment settlement

22.1 Inline settlement path

ADR 0047 returns classification to `Strategy`, following the upstream Liquid Signal boundary:

```

appliedBribeBps ← Resonance.bribeBps()           -- before payment-token interaction
paymentToken.safeTransferFrom(buyer, Strategy, p)
bribeAmount ← mulDiv(p, appliedBribeBps, 10000)
fundAmount ← p - bribeAmount
if fundAmount ≠ 0: paymentToken.safeTransfer(Fund, fundAmount)
if bribeAmount ≠ 0: paymentToken.safeTransfer(BribeRouter, bribeAmount)

```

The Strategy holds immutable Fund and payment-token addresses. It sends no part of the acquired payment to Resonance, a fee recipient, or a caller-selected destination. A failed Fund or Router transfer reverts the complete purchase; there is no deferred Fund settlement or liability ledger.

22.2 Bounded prospective classification

`Resonance` holds the only rate:

```

uint256 public constant BPS = 10_000;
uint256 public constant DEFAULT_BRIBE_BPS = 1_000;
uint256 public constant MAX_BRIBE_BPS = 2_000;
uint256 public bribeBps = DEFAULT_BRIBE_BPS;

```

The Strategy snapshots `bribeBps` before `safeTransferFrom`, so even a callback-capable payment token cannot change the classification of the purchase whose transfer triggered the callback. The owner can change only later purchases. Because every admissible rate is at most 2,000, Fund receives at least 80% of **each** purchase under integer arithmetic. `Math.mulDiv` supplies a full-width product for the floored Bribe calculation.

This split applies to the **acquired payment asset**. It does not split USDG: Resonance still pays 100% of a Strategy's earned USDG to that Strategy.

DEFAULT ACQUISITION CLASSIFICATION

90%

10%

Fund

treasury backing for every GBX holder

Signalers

reward for signaling this Strategy

Each purchase floors its Bribe share; Strategy sends the complement directly to Fund.

THE DEFAULT IS 90% FUND / 10% BRIBE. STRATEGY FLOORS EACH PURCHASE'S BRIBE SHARE INDEPENDENTLY, SENDS FUND'S COMPLEMENT DIRECTLY, AND BUFFERS ONLY THE BRIBE SHARE.

22.3 Per-purchase floors and partition dependence

Formula F-21 (per-purchase classification). For each payment a_i at its captured rate r_i :

```

Bribe_i = [a_i · r_i / BPS]
Fund_i  = a_i - Bribe_i

cumulative Bribe = Σ_i [a_i · r_i / BPS]
cumulative Fund  = Σ_i a_i - cumulative Bribe

```

There is no `splitRemainder`, so in general $\sum_i [a_i \cdot r_i / BPS] \neq [\sum_i (a_i \cdot r_i) / BPS]$. Partitioning can change cumulative classification by sub-token units.

PARTITION-DEPENDENT PER-PURCHASE FLOORS

The same ten raw units can classify differently at the same 10% rate.

ONE 10-UNIT PURCHASE

Fund 9

1

TEN 1-UNIT PURCHASES

1

1

1

1

1

1

1

1

1

1

Fund 10 · Bribe 0

No cross-purchase remainder is stored. The smaller state machine is deliberate; the partition-dependent rounding difference is accepted.

PAYMENT CLASSIFICATION IS INTENTIONALLY PARTITION-DEPENDENT: ONE TEN-UNIT PURCHASE YIELDS ONE BRIBE UNIT, WHILE TEN ONE-UNIT PURCHASES YIELD NONE.

Minimal example at 10%. One payment of ten raw units yields Fund 9 / Bribe 1. Ten separate one-unit purchases each floor the Bribe share to zero, yielding Fund 10 / Bribe 0. ADR 0047 accepts this result in exchange for removing cross-purchase carry state. The invariant is per-purchase conservation and the 80% Fund floor, not frequency independence.

22.4 Minimal BribeRouter buffer

`BribeRouter` has only immutable `paymentToken` and `bribe` references. Strategy sends only the Bribe share there; the Router does not know Fund, read the global rate, classify payments, or record liabilities. Its only action is permissionless `route()` :

```
amount ← paymentToken.balanceOf(router)
if amount = 0: return 0
if amount < Bribe.REWARD_DURATION(): return 0
if amount < Bribe.remainingReward(paymentToken): return 0
approve Bribe for amount
Bribe.notifyReward(paymentToken, amount)
emit RewardRouted(bribe, paymentToken, amount)
```

The complete balance is used. Compatible direct donations are therefore indistinguishable from Strategy-provided shares and join the next notification. A below-threshold call is a no-op. If Bribe notification reverts, the transaction reverts and the buffer remains. A temporary token failure can be retried if it clears; lifetime-cap exhaustion is permanent and strands the buffer. In either case, the earlier Strategy purchase is unaffected because notification is a separate transaction.

The Strategy payment token is registered as reward slot 1 of 16 when the Strategy graph is created (§20.1).

22.5 Settlement identities and non-identities

For a successful purchase using a supported standard token:

```
p = fundAmount + bribeAmount
fundAmount = p - ⌊p·r/BPS⌋
bribeAmount = ⌊p·r/BPS⌋
r ≤ 2000 ⇒ fundAmount ≥ p - ⌊p/5⌋
```

These are arithmetic identities evaluated before transfer. The Router's complete token balance is its buffer; there is no distinction between accounted payment and donation, no Fund or Bribe liability, and no `paymentSurplus` view. No cross-purchase conservation identity is asserted (§22.3), and unsupported token mechanics can violate the assumed movement semantics (§36).

22.6 GBX-denominated Strategies

A Strategy whose payment token is GBX has no special settlement branch. Its Fund complement travels `buyer` → `Strategy` → `Fund` and remains supply-neutral until anyone calls permissionless `Fund.burnGBX`. Its nonzero Bribe share travels `buyer` → `Strategy` → `BribeRouter`, then is streamed as GBX to that Strategy's signalers once the buffer qualifies.

Operational consequence. Fund-held GBX remains in current `gbx.totalSupply()` and therefore in the `effectiveTotalSupply()` redemption denominator (§25.2) until burned. A redeemer should burn accumulated Fund GBX before quoting or executing a redemption.

23. Bribe reward accounting

`Bribe` is a bounded multi-token adaptation of the Synthetix `MultiRewards` cumulative-index engine. ADR 0047 removes the protocol-specific queue, pause, carry, Fund-liability, and exact-transfer layers; the remaining differences from scalar Resonance are the token registry, per-token state, lifetime cap, and claim loop.

Constants: $\text{REWARD_DURATION} = 7 \text{ days}$, $\text{REWARD_PRECISION} = 10^{36}$, $\text{MAX_REWARD_TOKENS} = 16$.

23.1 State inventory

STATE	MEANING
<code>totalSignalWeight</code> , <code>signalWeightOf[a]</code>	Resonance-controlled virtual signal supply and account balances
<code>_rewardTokens</code> , <code>isRewardToken[t]</code>	Append-only registry, capped at sixteen
<code>rewardData[t]</code>	<code>periodFinish</code> , <code>rewardRate</code> , <code>lastUpdateTime</code> , stored <code>rewardPerSignal</code>
<code>accountRewardPerSignalPaid[a][t]</code>	Index already incorporated for one account and token
<code>rewards[a][t]</code>	Accrued whole-token amount payable only to <code>a</code>
<code>lifetimeRewardNotified[t]</code>	Monotonic cumulative raw units admitted for <code>t</code>

23.2 Stream mathematics

For registered token `t`, let $T = \min(\text{now}, \text{periodFinish}[t])$, $P = 10^{36}$, and $W = \text{totalSignalWeight}$.

Formula F-16 (cumulative reward per signal).

$$\begin{aligned} \text{rewardPerSignal}(t) &= \text{stored}[t] && \text{if } W = 0 \\ &= \text{stored}[t] + [(T - \text{lastUpdateTime}[t]) \\ &\quad \cdot \text{rewardRate}[t] \cdot P / W] && \text{otherwise} \end{aligned}$$

Formula F-17 (account entitlement).

$$\begin{aligned} \text{earned}(a, t) &= \text{rewards}[a][t] \\ &+ [\text{signalWeightOf}[a] \cdot (\text{rewardPerSignal}(t) - \text{paid}[a][t]) / P] \end{aligned}$$

`_updateReward(a, t)` stores the current index and applicable timestamp, then stores `earned(a, t)` and the new paid index when $a \neq 0$. `_updateAllRewards(a)` repeats that fixed work for every registered token before a signal-balance change or all-token claim.

23.3 Notification and leftover rollover

`notifyReward(t, amount)` is permissionless for a registered token and executes in this order:

Formula F-18 (Bribe restart).

```
require amount ≥ REWARD_DURATION
require lifetimeRewardNotified[t] + amount ≤ MAX_LIFETIME_REWARD_AMOUNT
remaining ← remainingReward(t)
require amount ≥ remaining
checkpoint token t globally
safeTransferFrom(caller, Bribe, amount)
rewardRate[t] ← [(amount + remaining) / REWARD_DURATION]
lastUpdateTime[t] ← now
periodFinish[t] ← now + REWARD_DURATION
lifetimeRewardNotified[t] += amount
```

This is ordinary Synthetix leftover rollover: a valid top-up combines the new amount with whole units scheduled at the old rate and restarts from `now`. Equality with `remainingReward` is accepted. The duration floor guarantees a nonzero new rate; the leftover gate makes repeatedly restarting a live stream proportionately expensive.

There is **no queue**. A valid notification always starts or restarts immediately.

23.4 Zero supply and accepted surplus

Reward time does **not pause** when `totalSignalWeight = 0`. `rewardPerSignal` holds the index constant, while a checkpoint advances `lastUpdateTime` to the applicable timestamp. Emission from that zero-supply interval is never allocated to a later entrant.

For each token, the following remain unallocated Bribe balance rather than being carried or sent to Fund:

1. `(amount + remaining) mod REWARD_DURATION` at schedule creation;
2. the global-index floor in F-16;
3. the account floor in F-17;
4. emission while signal supply is zero; and
5. direct transfers that did not enter through `notifyReward`.

There are no global, account, or Fund carry buckets; no Fund reward liability; and no exact conservation identity. Checkpoint and account-partition frequency can affect the accumulated surplus. ADR 0047 explicitly accepts this in exchange for retaining the small upstream-shaped state machine.

23.5 Bounded reward registry

`MAX_REWARD_TOKENS = 16`, append-only, `onlyResonance`. The cap is what makes every mandatory loop — `_updateAllRewards`, signal-balance checkpoints, and the all-token claim — constant-bounded (finding A-08). The higher bound accepts approximately doubled worst-case loop work in exchange for fifteen independent incentive slots after the automatic payment token. The scalar-addition, withdrawal, and composed-move measurements recorded for ADR 0048 predate ADR 0051 and do not establish batch headroom. Current maximum-bound tests must measure several Strategies with sixteen reward streams each. Batch length is caller-controlled, while scalar removal remains the bounded exit. Historical numbers remain local engineering evidence only.

23.6 Lifetime notification cap (ADRs 0035 and 0037)

Each `(Bribe, token)` pair carries a monotonic counter of every raw unit ever admitted through `notifyReward`, whether the notifier is a `BribeRouter` settling the bounded automatic share (10% by default) or an independent funder. The immutable ceiling is:

Formula F-24.

$$P = \text{REWARD_PRECISION} = 1e36$$

$$\text{MAX_LIFETIME_REWARD_AMOUNT} = \lfloor (2^{256} - 1) / P \rfloor$$

A notification is rejected with `RewardLifetimeCapExceeded(token, notified, requested, maximum)` when

$$\text{amount} > \text{MAX_LIFETIME_REWARD_AMOUNT} - \text{lifetimeRewardNotified}[t]$$

The check runs **before** the active-leftover test, checkpoint, or token interaction, so a rejection leaves the caller's balance and every schedule untouched.

Why the counter is monotonic. Claims, stream completion, Strategy death, and zero-supply intervals do not reduce the cumulative reward-per-signal index already written to storage. Reopening capacity after payout would therefore allow repeated notifications to exhaust that index. Because every signal addition and removal checkpoints all registered tokens, such an overflow could strand escrowed GBX. This was finding **BR-2**.

Safety argument. One admitted raw unit contributes at most P scaled units to the global index, and the smallest possible virtual supply is one raw signal unit, which assigns the whole scaled amount. With lifetime notifications N :

$$\text{rewardPerSignalStored} \leq N \cdot P \quad \text{and} \quad N \leq \lfloor (2^{256} - 1) / P \rfloor$$

so the stored and previewed index remain representable. A one-unit virtual supply attains the bound, making this the largest history-independent limit that is safe under arbitrary supply changes.

Consequences. The cap is approximately 1.158×10^{41} raw units, or 1.158×10^{23} whole units for an 18-decimal token, and constrains no conventional asset; it can bind an unusually high-decimal token. Reaching it blocks only new notifications for that one token in that one Bribe — existing rewards, claims, and signal removals continue (§32, L-9). If an automatic Strategy-payment notification is rejected, the complete balance remains buffered in `BribeRouter`; the Fund share was already transferred during the purchase (§22.4). Direct Bribe donations never enter the index and never consume the cap. No epoch reset, retirement withdrawal, or rescue path is introduced.

23.7 Claim isolation

Two claim shapes remain: `claimRewards(account)` checkpoints and attempts every registered token, while `claimReward(account, token)` checkpoints and pays exactly one registered token. Both pay the entitled `account`, never `msg.sender`. The scalar claim is what lets a signaler route around a broken reward token; caller-selected batch claims are periphery rather than core.

23.8 Exit liveness

`Bribe.removeSignalWeight` performs **only** reward checkpoints and virtual-balance decrements. It contains no `transfer`, `transferFrom`, or `safeTransfer`. Therefore signal removal does not interact with a reward token and cannot fail because that token is frozen — design goal G4. Verified by `invariant_EveryActorCanFullyWithdrawSignals`.

24. Fund custody

`Fund` is `ReentrancyGuard` only. It is **not** `Ownable`, has no roles, and its only storage is the immutable `gbx`.

24.1 Value exits

Exactly two, both permissionless:

FUNCTION	EFFECT
<code>burnGBX(amount)</code>	Burns <code>amount</code> of Fund's own GBX balance
<code>redeem(...)</code>	Burns caller's GBX, transfers floored pro-rata basket

There is no sweep, rescue, recovery, migration, or administrative withdrawal of any kind. Assets omitted by every redeemer remain in Fund indefinitely, backing the residual supply.

24.2 Registry-free design

Fund maintains **no asset list**. Any ERC-20 transferred to it becomes redeemable backing without review, registration, or governance action.

Therefore: presence in Fund does **not** indicate protocol endorsement. Official protocol membership is represented solely by Strategies registered in `Resonance`. Interfaces must label unsolicited Fund balances separately, support manual asset-address entry, and warn that omissions are permanently forfeited.

25. GBX redemption

25.1 Interface

```
function redeem(uint256 gbxAmount, address receiver, address[] calldata tokens) external nonReentrant
```

Validation: `gbxAmount ≠ 0`; `receiver ∉ {0, address(this)}`; `tokens.length ≠ 0`; every entry unique and not GBX or zero (§25.4).

25.2 Algorithm

```
1. require gbx.minterLocked() ∧ mine.code.length > 0 ∧ IMine(mine).gbx() = gbx
2. supplyBeforeBurn ← IMine(mine).effectiveTotalSupply() -- minted plus all pending mining issuance
3. require supplyBeforeBurn ≠ 0 ∧ gbxAmount ≤ supplyBeforeBurn
4. for each token i:
    _markToken(i) -- transient duplicate guard
    balancesBefore[i] ← IERC20(i).balanceOf(Fund)
    payouts[i] ← mulDiv(balancesBefore[i], gbxAmount, supplyBeforeBurn)
5. transferFrom(msg.sender → Fund, gbxAmount) ; gbx.burn(gbxAmount)
6. for each token i:
    require IERC20(i).balanceOf(Fund) ≥ balancesBefore[i] -- pre-transfer guard
    if payouts[i] ≠ 0: transfer exactly payouts[i] to receiver (exact-delta checked)
    _clearToken(i)
7. for each token i:
    require IERC20(i).balanceOf(Fund) ≥ balancesBefore[i] - payouts[i] -- final basket guard
```

Formula F-19 (payout).

```
payout_i = [ balanceOf_i(Fund) · gbxAmount / supplyBeforeBurn ]
```

Units. Raw units of token `i`. *Rounding.* Floor — the residue stays in Fund for the residual supply, so rounding always favors remaining holders. *Overflow.* `mulDiv` at 512-bit intermediate precision.

Every payout uses the **same** `supplyBeforeBurn`, so a multi-asset basket is internally consistent. The entire operation is atomic: any revert unwinds the burn and all transfers.

25.3 Why effective supply includes pending mining

Mining accrual is lazy (§12.6): a miner's earned GBX is unminted until its slot tenure is replaced. Reading only `totalSupply()` would exclude already-earned issuance, and a redeemer would receive a share computed against an artificially small denominator — diluting miners in favor of redeemers. `effectiveTotalSupply()` eliminates the timing advantage in constant time and does not mutate `Mine`. This is the remediation of finding **A-10**.

25.4 Duplicate detection via EIP-1153

```
slot = keccak256(abi.encode(REDEMPTION_NAMESPACE, token))
_markToken: reject token ∈ {0, gbx}; if tload(slot) ≠ 0 revert DuplicateToken; tstore(slot, 1)
_clearToken: tstore(slot, 0)
```

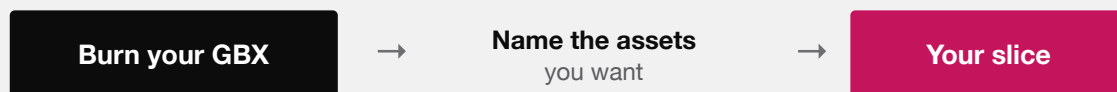
This gives $O(n)$ duplicate detection with **no** permanent storage writes, no sorting requirement, no asset registry, no token IDs, and no monotonic nonce. Marks are cleared after each successful payout so that a second redemption later in the same transaction is fully independent.

Deployment prerequisite. The target chain must support EIP-1153 (Cancun). This is a hard requirement recorded in `docs/TRUST_ASSUMPTIONS.md`.

25.5 The shared-ledger guard

Steps 6 and 7 defeat an attack in which two distinct addresses front the same underlying balance ledger. Without the final pass, a redeemer could nominate both facades and extract the same backing twice. The final pass requires each selected address to retain at least `balancesBefore[i] - payouts[i]`, catching asymmetric aliases where only the later transfer mutates the earlier address's reported balance. This is the remediation of finding **E-01**, evidenced by `test_RedeemRejectsDifferentAddressesThatDebitOneSharedLedger` and `test_RedeemFinalPassRejectsAnAsymmetricAliasSideEffect`.

REDEEMING



For each asset you name:

$\text{floor}(\text{Fund's balance} \times \text{your GBX} \div \text{total GBX supply})$

The supply is snapshotted after every miner is credited, so nobody is diluted by unminted rewards. Assets you leave out are forfeited — permanently, to everyone else.

It is one atomic step. If any transfer fails, the burn is undone too.

REDEMPTION IN FULL. NO QUEUE, NO GATE, NO DISCRETION — ONLY ARITHMETIC AGAINST A SNAPSHOT.

25.6 Worked redemption example

State. Fund holds 50 WBTC ($5 \cdot 10^9$ raw at 8 decimals) and 400 ETH ($4 \cdot 10^{20}$ raw at 18 decimals). `gbx.totalSupply() =  $10^8 \cdot 10^{18}$` . Miners have accrued $10^6 \cdot 10^{18}$ unminted GBX. Redeemer burns `gbxAmount =  $2.5 \cdot 10^5 \cdot 10^{18}$` (250,000 GBX).

Step 2. `effectiveTotalSupply()` returns $1.01 \cdot 10^{26}$ without minting or changing a slot.

Step 4.

```
WBTC: [ 5 · 10^9 · 2.5 · 10^23 / 1.01 · 10^26 ] = [ 12_376_237.62 ] = 12_376_237 raw = 0.12376237 WBTC
ETH : [ 4 · 10^20 · 2.5 · 10^23 / 1.01 · 10^26 ] = 990_099_009_900_990_099 raw ≈ 0.990099009900990099 ETH
```

Step 5. Supply falls to $1.0075 \cdot 10^{26}$ (100,750,000 GBX).

Counterfactual. Without step 2, `supplyBeforeBurn = 10^26` and the WBTC payout would be `12_500_000` raw (0.125 WBTC). The checkpoint costs this redeemer `123_763` raw WBTC — exactly the dilution attributable to the miners’ already-earned claim, and therefore correct.

This reproduces the mechanism modelled in `packages/simulations/fixtures/economic-scenarios.json`, whose analogous scenario yields `495_049_504_950` raw USDG with the checkpoint versus `500_000_000_000` without.

26. External LP Strategy

ADR 0050 removes canonical protocol-owned liquidity. A reviewed, externally created fungible Uniswap v2-style USDG/GBX LP token may be registered during bootstrap as one ordinary Strategy payment token.

26.1 Ordinary Strategy treatment

The LP-token Strategy uses the same bounded descending-price acquisition, signaling, irreversible kill behavior, and global prospective Fund/Bribe split as every other Strategy. The LP token received from a buyer is transferred to Fund and the paired Bribe path in the same 80%-to-100% / 0%-to-20% classification. No per-Strategy exception exists, and no LP token address is hard-coded.

26.2 Core boundary

No core component creates, seeds, owns, custodies, prices, rebalances, compounds, harvests, or swaps liquidity. There is no canonical position, liquidity NFT, liquidity owner, or protocol fee-harvest path. Registering an external LP token neither proves its safety nor guarantees a liquid GBX market.

26.3 Deployment responsibility

The exact LP token address, pair identity, venue, underlying token identities, and code provenance are reviewed deployment inputs. Registration is ordinary `Resonance.addStrategy` bootstrap policy. If no suitable token is reviewed, the deployment must not invent an address or claim guaranteed liquidity.

27. Ownership lifecycle and the enforcement boundary

§15 specifies the owner-gated surface. This section specifies **who holds that owner role over time**, and draws the line between what Solidity enforces and what a deployment must prove.

27.1 Ownership at each stage

STAGE	RESONANCE.OWNER ()	CAPABILITY HELD
Construction	<code>initialOwner</code> constructor argument	All owner-gated calls, including <code>setResonanceRouter</code>
Bootstrap	Temporary deployment setup owner	Binds the Router; creates reviewed initial Strategies
After handoff (required)	Exact reviewed external executor	<code>addStrategy</code> , <code>killStrategy</code> , <code>addBribeRewardToken</code> , bounded prospective <code>setBribeBps</code>
Optional terminal state	<code>address(0)</code> after <code>renounceOwnership</code>	None; Strategy membership is frozen permanently

`SignalGBX`, `StrategyFactory`, and `BribeFactory` each retain a nominal `Ownable` owner after their one-time `setResonance` binding is consumed, but no remaining function is gated on it (§9.1). `Mine`, `Fund`, `Strategy` and `BribeRouter` are not `Ownable` at all. `Bribe` gates `addRewardToken` on the immutable `resonance` address rather than on an owner.

27.2 What Solidity enforces, and what it does not

This distinction is the single most important qualification in this document.

Enforced by the contracts, unconditionally and without reference to any deployment procedure:

- `GBX.setMinter` succeeds at most once and permanently sets `minterLocked` (§11).
- `Mine` has no owner and no capacity-changing function; `SLOT_COUNT` is `constant` (§12).
- `Resonance.setResonanceRouter`, `SignalGBX.setResonance`, `StrategyFactory.setResonance`, and `BribeFactory.setResonance` each revert after their first success.
- One-time bindings validate reciprocal identity — a `Mine` must report the same `GBX`, a `Router` the same `Resonance` and `USDG`, a `Resonance` the same `SignalGBX` and factory pair (§28.1).
- `killStrategy` reverts on the final live `Strategy`.
- `Fund`, `Strategy`, and `BribeRouter` expose no owner, sweep, rescue, or migration path.

Not enforced by any contract in this repository, and therefore a deployment obligation that must be proven by signed evidence rather than asserted:

- That the `Resonance` owner after handoff is the intended external governance executor rather than an EOA, a compromised address, or a lookalike contract.
- That the temporary setup owner retains no authority afterward.
- That the `Mine` bytecode contains the reviewed initial rate, provisional halving period, tail rate, price multiplier, and minimum initial price.
- That the deployed dependency is the canonical `USDG` contract on the target chain, and that any bootstrap external LP `Strategy` uses the exact reviewed fungible token address.
- That `Mine`'s immutable `USDG` and `Router` form the intended pair; ADR 0045 makes this a pinned post-deployment evidence obligation rather than a `Mine` constructor invariant.

Reciprocal binding checks plus ADR 0045's post-deployment `Mine/Router` verification reject a *crossed* protocol graph. They cannot distinguish a malicious lookalike that returns the expected identities, and the protocol has no upgrade, successor, or migration authority with which to repair a wrong value after exposure. This is finding **M-03** — an open High release gate — and the residue of **E-02**.

Where any repository document states an ownership or role condition as an “invariant”, it is correctly read as a **deployment obligation**. This was recorded as discrepancy D-5 (§43).

27.3 Handoff sequence and evidence

Ownership closure is step 8 of `docs/DEPLOYMENT.md`, and it is gated on step 7: deployment stops unless a later ADR has selected and reviewed the external governance integration. The sequence is:

1. Setup owner binds `ResonanceRouter` (single-use).
2. Setup owner creates every reviewed bootstrap `Strategy` and registers reviewed `Bribe` reward tokens.
3. A later ADR pins the external governance provider, release, bytecode, permission graph, and voting semantics.
4. Setup owner renounces ownership of `SignalGBX`, `StrategyFactory`, and `BribeFactory` after their bindings are consumed.
5. Setup owner calls ``transferOwnership(exact reviewed executor)`` on `Resonance`. No intermediate custodian.
6. Verify ``Resonance.owner()``, the handoff receipt, all three renunciations, and that the coordinator retains no authority.

Bootstrap Strategies are created **before** handoff deliberately: the initial membership is part of the reviewed deployment rather than the first act of an unreviewed governance system. The consequence is that the setup owner's key retains Resonance authority until step 5 completes, and a deployment interrupted between steps 2 and 5 leaves a protocol with a live Resonance admin key. Until step 4, it also owns the three setup shells. There is no contract-level timeout, escrow, or forced handoff.

28. Deployment and immutable bindings

28.1 Binding table

BINDING	GUARD	RECIPROCAL CHECK	REUSABLE?
<code>GBX.setMinter(Mine)</code>	<code>msg.sender = minter , ~minterLocked</code>	<code>IMine(m).gbx() = GBX</code>	No
<code>SignalGBX.setResonance(R)</code>	<code>onlyOwner , unset</code>	<code>R.signalGBX() = SignalGBX</code>	No
<code>StrategyFactory.setResonance</code>	<code>onlyOwner , unset</code>	<code>R.strategyFactory() = this</code>	No
<code>BribeFactory.setResonance</code>	<code>onlyOwner , unset</code>	<code>R.bribeFactory() = this</code>	No
<code>Resonance.setResonanceRouter</code>	<code>onlyOwner , unset</code>	<code>Router.resonance() = this and Router.usdg() = usdg</code>	No
Bribe constructor	nonzero contract address	immutable Resonance caller identity; no Fund dependency	n/a

Every listed one-time binding read is guarded and reverts on failure or on a mismatched identity. ADR 0045 deliberately moves verification of Mine's configured Router address and both contracts' canonical USDG identity out of Mine construction and into pinned deployment evidence; a mismatched Mine candidate must be abandoned before GBX binding or exposure.

`SignalGBX` additionally refuses **all** signal additions until its Resonance binding completes (`ResonanceNotSet`), so no user can deposit into a partially-wired graph.

28.2 What reciprocal checks do and do not prove

The onchain checks prove **consistency** for the bindings they guard: contract A and contract B agree they refer to each other, and to the same USDG or GBX. Pinned deployment reads provide the equivalent consistency evidence for Mine's Router/token pairing. Neither mechanism proves **honesty**: a malicious lookalike that returns the expected identities passes every check.

This materially reduces accidental cross-wiring but does not close finding **M-03**, which remains an open High release gate requiring exact runtime code hashes, constructor arguments, transaction receipts, and a signed manifest.

28.3 Deployment order (summary)

The intended sequence from `docs/DEPLOYMENT.md`: deploy GBX with a temporary coordinator as minter → deploy Fund, SignalGBX, both factories → deploy Resonance with a temporary setup owner, bind it into SignalGBX and both factories, deploy ResonanceRouter and bind it → deploy Mine and verify its identities → `GBX.setMinter(Mine)` **irreversibly** → create every reviewed bootstrap Strategy, including the reviewed external LP token if selected, while the setup owner still controls Resonance → **stop** unless a later ADR has selected and reviewed the external governance integration → transfer Resonance ownership directly to the exact reviewed external executor and prove the coordinator retains no authority → reconcile all runtime bytecode, arguments, bindings, ownership, and custody; also renounce the consumed SignalGBX, StrategyFactory, and BribeFactory setup shells and verify their owners are zero (§27.3).

28.4 The irreversibility budget

Every one of the following is permanent and unrepairable once executed:

ACTION	FAILURE MODE IF WRONG
GBX.setMinter	Wrong or malicious issuer forever; no second minter possible
Any setResonance / router binding	Permanently crossed graph
Mine bytecode economics	Wrong emission curve forever
Resonance ownership handoff	Wrong or hostile administrator for the four capabilities
Retaining the temporary setup owner	A live admin key in a protocol that claims to have none
Bootstrap Strategy set	Unwanted Strategies exist forever (killable, not removable)

A failed setup must be abandoned entirely before use. There is no repair authority.

29. State machines

29.1 Mining slot

FROM	TRANSITION	TO	EFFECTS
Empty	mine at price p	Occupied	100% of $p \rightarrow$ Router deposit; $tps \leftarrow \lfloor \text{globalTps}/16 \rfloor$; epochId++
Occupied	mine at price $p > 0$	Occupied	settle this slot; $\lfloor 0.8p \rfloor \rightarrow$ claim, $\lfloor 0.2p \rfloor \rightarrow$ router; assign new tps
Occupied	mine at price $p = 0$	Occupied	settle this slot; no token movement ; assign new tps ; epochId++
Occupied	Fund. redeem	Occupied	no slot mutation; effective supply includes the slot's pending emission

A slot never returns to Empty. The permanent slot count is sixteen.

29.2 Signal position (account $\times$ Strategy)

FROM	TRANSITION	TO	GUARD
None	addSignal / addSignalMany	Signalling	every destination live; caller holds GBX and allowance
Signalling	addSignal / addSignalMany	Signalling (+x)	every destination live; aggregate custody succeeds
Signalling	removeSignal / removeSignalMany	Signalling (-x)	each amount fits its paired-Bribe account balance
Signalling on dead	removeSignal / removeSignalMany	Signalling (-x)	permitted; active weight unchanged
Signalling on dead	addSignal / addSignalMany	—	reverts StrategyAlreadyDead

Because of Identity I-4 there is no None $\rightarrow$ Idle transition and no idle state at all: a position either exists on some Strategy or the sGBX does not exist.

29.3 Strategy

FROM	TRANSITION	TO	NOTES
—	<code>addStrategy</code>	Live	Deploys Strategy + BribeRouter + Bribe; <code>++liveStrategyCount</code> ; index initialized to current
Live	<code>killStrategy</code> when <code>liveStrategyCount > 1</code>	Dead	Checkpoints first; weight excluded; <code>--liveStrategyCount</code> ; irreversible
Live	<code>killStrategy</code> when <code>liveStrategyCount = 1</code>	Live	reverts <code>FinalLiveStrategy</code> — a replacement must be added first
Dead	—	Dead	No revive transition exists

The final-live-Strategy guard (ADR 0031, superseding ADR 0029's permission to kill it) guarantees a valid signal destination always exists — necessary now that signaling is the only way to hold sGBX. The owner replaces the last Strategy by calling `addStrategy(replacement)` before `killStrategy(previous)`; whether those two calls can be atomically batched is a property of the external governance system, not of the core. Verified by `test_KillingTheFinalLiveStrategyRevertsAfterBootstrap`.

29.4 Strategy auction epoch

FROM	TRANSITION	TO	EFFECTS
Epoch n	<code>buy</code> at price <code>p > 0</code>	Epoch n+1	snapshot rate → <code>distributeRevenue</code> → pull <code>p</code> → direct split → pay USDG → <code>reprice</code>
Epoch n	<code>buy</code> at price <code>p = 0</code>	Epoch n+1	snapshot rate → <code>distributeRevenue</code> → no payment → pay USDG → <code>p_min</code> restart
Epoch n	<code>buy</code> with zero USDG	Epoch n	reverts <code>EmptyRevenue</code>
Epoch n	<code>buy</code> with stale <code>epochId</code>	Epoch n	reverts <code>EpochIdMismatch</code>

29.5 Bribe reward stream (per token)

FROM	TRANSITION	TO	EFFECTS
Inactive	<code>valid</code> <code>notifyReward(token, amount)</code>	Active	floor rate; start seven days at <code>now</code>
Active	<code>notifyReward</code> at <code>amount ≥ REWARD_DURATION</code> and <code>≥ remainingReward</code>	Active	checkpoint; combine ordinary leftover; restart seven days
Active	notification below either gate	Active	reverts without changing schedule or pulling tokens
Active	<code>totalSignalWeight → 0</code>	Active	schedule time continues; later checkpoint leaves interval surplus
Active	<code>now ≥ periodFinish</code>	Inactive	index preserved; unscheduled balance remains surplus

There is no queued or paused state. A dead Strategy with no remaining signal supply is economically terminal for future allocation, but the Bribe does not encode a separate terminal state.

29.6 BribeRouter buffer

FROM	TRANSITION	TO	NOTES
Empty	Strategy share or direct donation	Below/qualifying	Token balance itself is the only state
Empty	<code>route()</code>	Empty	Returns zero
Below threshold	<code>route()</code>	Below threshold	Returns zero; needs more balance, or lower <code>remainingReward</code> when that gate alone binds
Qualifying	<code>route()</code> succeeds	Empty	Complete balance enters Bribe stream
Qualifying	Bribe notification reverts	Qualifying	Entire call reverts; complete buffer remains

29.7 Resonance ownership

FROM	TRANSITION	TO	NOTES
Constructor owner	Deployment bootstrap (§27.3)	Setup owner	Binds Router; creates reviewed bootstrap Strategies
Setup owner	<code>transferOwnership(executor)</code>	External executor	Required before any user funds; step 8 of deployment
Any owner	<code>transferOwnership(other)</code>	Other owner	Unconstrained by the core
Any owner	<code>renounceOwnership()</code>	<code>address(0)</code>	Irreversible ; Strategy membership frozen permanently

The core imposes no delay, approval, confirmation, or two-step acceptance on any of these transitions. There is no state in which an ownership change is pending and observable before it takes effect.

30. Accounting identities

Only identities the implementation can actually prove are asserted. Where exact conservation does not hold, an inequality is stated instead and the residue is named.

I-1 — GBX supply. `totalSupply = lifetimeMinted - lifetimeBurned`. *Exact.* (§11.3)

I-2 — Mine solvency. `USDG.balanceOf(Mine) ≥ totalClaimableMinerPayments`, equality absent donations. *Exact modulo donations.* (§12.4)

I-3 — sGBX collateralization. `sGBX.totalSupply ≤ GBX.balanceOf(SignalGBX)`, equality absent donations. *Exact modulo donations.* (§13.2)

I-4 — Mandatory signal-backing. $\forall a: \text{sGBX.balanceOf}(a) = \sum_s \text{Bribe}(s).\text{signalWeightOf}(a)$, and `sGBX.totalSupply() =  $\sum_s \text{Bribe}(s).\text{totalSignalWeight}()$` , across live and killed Strategies. *Exact.* (§13.3)

I-5 — Per-purchase Strategy split. For every successful supported-token purchase at price `p` and captured rate `r`, $p = (p - \lfloor p \cdot r / \text{BPS} \rfloor) + \lfloor p \cdot r / \text{BPS} \rfloor$, with the first term transferred directly to Fund and the second transferred to BribeRouter. *Exact arithmetic; movement relies on the standard-token assumption.* (§22.5)

I-6 — Signal ledger consistency.

$\forall a: \sum_s \text{Bribe}(s).\text{signalWeightOf}(a)$	$= \text{SignalGBX.balanceOf}(a)$
$\forall s: \sum_a \text{Bribe}(s).\text{signalWeightOf}(a)$	$= \text{Bribe}(s).\text{totalSignalWeight}$
$\sum_{\{s \text{ live}\}} \text{Bribe}(s).\text{totalSignalWeight}$	$= \text{Resonance.totalSignalWeight}$

Exact. Note the third line ranges over **live** Strategies only; a killed Strategy retains its recorded balances while contributing zero to the active total. Verified by `invariant_AccountWeightsSumToAllRecordedStrategyWeight`, `invariant_BribeBalancesMirrorAccountSignals`, `invariant_BribeSupplyMirrorsStrategyWeight`, `invariant_StrategyWeightsSumToTheGlobalTotal`, and `invariant_DeadStrategiesAreExcludedFromActiveWeight`.

I-7 — Resonance solvency (inequality only).

$$\text{USDG.balanceOf(Resonance)} \geq \text{remainingRevenue()} + \sum_s \text{earnedRevenue}(s)$$

The nonnegative difference comprises rate, global-index, and per-Strategy floors, zero-active-weight emission, and direct donations.

This is deliberately not an equality with a bounded residue. No exact conservation and no lifetime dust bound is claimed for Resonance (§17.5). Verified as an inequality by `invariant_ResonanceIsSolventAgainstClaimableRevenue`, `invariant_ResonanceScheduledAndEarnedRevenueIsSolvent`, and `testFuzz_AccruedAndScheduledRevenueNeverExceedsTheHeldBalance`.

I-8 — Bribe solvency (inequality only). Under supported standard-token behavior, for each registered token t :

$$\text{IERC20}(t).\text{balanceOf(Bribe)} \geq \text{remainingReward}(t) + \sum_a \text{earned}(a, t)$$

The nonnegative difference includes schedule-rate, global-index, account, and zero-supply floors plus direct donations. No exact conservation identity or `rewardSurplus` accounting view exists after ADR 0047. Verified by the current scalar claim and custody fuzz properties.

I-9 — Fund redemption conservation. For each selected token i :

$$\begin{aligned} \text{balanceAfter}_i &= \text{balanceBefore}_i - \lfloor \text{balanceBefore}_i \cdot \text{gbxAmount} / \text{supplyBeforeBurn} \rfloor \\ \text{supplyAfter} &= \text{supplyBeforeBurn} - \text{gbxAmount} \end{aligned}$$

and therefore backing per remaining GBX is non-decreasing:

$$\text{balanceAfter}_i / \text{supplyAfter} \geq \text{balanceBefore}_i / \text{supplyBeforeBurn}$$

Exact, with the inequality strict whenever the floor discards a residue. Verified by `testFuzz_PayoutIsExactlyTheFloored-ProRataShare`, `testFuzz_BackupPerGBXNeverDecreasesOnRedemption`, `testFuzz_SequentialRedemptionsStaySolvent`, and `invariant_FundNeverOwesMoreThanItHolds`.

I-10 — Effective supply. `Mine.effectiveTotalSupply() = GBX.totalSupply() + Mine.pendingEmission()` before any checkpoint. *Exact.* Verified by `invariant_EffectiveSupplyIncludesEveryPendingEmission`.

I-11 — USDG routing under the supported-token model. Mine requests transfer of the nominal protocol share into `ResonanceRouter` without measuring balance deltas. On a qualifying `route()`, the Router approves its complete observed balance and Resonance pulls the requested amount with `SafeERC20`. For standard non-rebasing USDG, each requested amount arrives; ADRs 0047 and 0049 remove duplicate balance-delta checks at these boundaries. Verified by `testFuzz_RoutingConservesTheCompleteBalance`.

30.1 Identities deliberately *not* asserted

TEMPTING CLAIM	WHY IT IS NOT ASSERTED
Resonance USDG is exactly conserved	Three floor/discard sources; see I-7 and §17.5
Resonance dust is bounded over the protocol lifetime	Depends on unbounded checkpoint frequency and lifetime
Bribe rewards are exactly conserved	Schedule, index, account, and zero-supply floors are accepted surplus (§23.4)
Every Bribe reward is eventually claimable	False after BR-1 abandonment (§20.4)
Fund backing per GBX is non-decreasing under all operations	Only proven for redemption; unsolicited transfers and burns move it in either direction
Aggregate GBX issuance $\leq$ current global rate	False while pre-halving tenure rates remain locked (M-01, §12.6)
The <code>Resonance</code> owner is the intended administrator	Procedural, not code-enforced (§27.2)

31. Security invariants

ID	INVARIANT	EVIDENCE
S-1	Only the permanently bound Mine can mint GBX, and only after <code>minterLocked</code>	<code>test_OnlyPermanentlyBoundMineCanMint</code>
S-2	<code>slot.tps</code> is never rewritten during a tenure	<code>test_TimeBasedHalvingNeverRepricesAnIncumbent</code>
S-3	Mine always has exactly 16 slots and no capacity mutation	<code>test_LaunchesWithSixteenEmptySlotsAndPermanentMiningAuthority</code>
S-4	sGBX is non-transferable in every path, including self and zero-value	<code>test_TransfersRemainPermanentlyDisabled</code>
S-5	Only SignalGBX may mutate signal state on Resonance	<code>test_OnlySignalGBXCanMutateAnotherAccountsSignal</code>
S-6	Only Resonance may mutate Bribe virtual balances or append reward tokens	<code>test_VirtualBalanceMutationIsResonanceOnly</code>
S-7	Only the bound Resonance may deploy through either factory	<code>test_FactoriesAreResonanceOnly</code>
S-8	Strategy snapshots the global Bribe rate before payment-token interaction	<code>test_PaymentTokenCallbackCannotRetroactivelyChangeTheCurrentPaymentsSnapshot</code>
S-9	Only the bound Router may call <code>Resonance.notifyRevenue</code>	<code>test_NotifyRevenueIsRouterOnlyAndRejectsZero</code>
S-10	A same-transaction signal cannot capture newly notified revenue or Bribe rewards	<code>test_FlashSignalWeightCannotRedirectANewNotification</code> , <code>test_FlashSignalWeightCannotStealAccruedBribeRewards</code>
S-11	A Bribe top-up below duration or active <code>remainingReward</code> cannot mutate the live stream	<code>test_NotifyRejectsUnregisteredAndBelowDurationAmounts</code> , <code>test_ActiveTopUpBelowTheAmountLeftRevertsWithoutChangingTheStream</code>

ID	INVARIANT	EVIDENCE
S-12	Bribe emission elapsed at zero supply cannot be captured by a later signaler	<code>test_ElapsedRewardsAtZeroSupplyRemainUnclaimableSurplus</code>
S-13	Each supported-token purchase sends its floored Fund complement inline	<code>test_ADustPaymentFloorsTheBribeShareAndGoesDirectlyToFund ,</code> <code>testFuzz_OnePurchaseUsesTheCurrentRateAndFloorsItsShare</code>
S-14	Redemption rejects GBX, zero, and duplicates in any position	<code>test_RedeemRejectsDuplicatesInAnyPosition</code>
S-15	A redemption basket cannot double-consume one shared backing ledger	<code>test_RedeemRejectsDifferentAddressesThatDebitOneSharedLedger</code>
S-16	Redemption is atomic: any failure reverts the burn and all transfers	<code>test_ASelectedFailingTransferRollsBackTheEntireRedemption</code>
S-17	Redemption includes all pending mining in a constant-time denominator	<code>test_RedemptionUsesEffectiveSupplyWithoutSettlingAnyMiner</code>
S-18	Reentrancy cannot double-claim a reward or double-fill an auction	<code>test_ReentrantRewardPayoutCannotDoubleClaim ,</code> <code>test_AHostilePaymentTokenCannotReenterTheSameStrategy</code>
S-19	GBX starts at zero supply	<code>test_ConstructorStartsWithZeroSupply</code>
S-20	No GBX can be minted before the permanent Mine binding	<code>test_OnlyPermanentlyBoundMineCanMint</code>
S-21	Each continuing administration call is owner-gated	<code>test_AddStrategyIsOwnerOnlyAndCreatesTheCompleteGraph ,</code> <code>test_KillStrategyIsOwnerOnlyPermanentAndBlocksNewSignal ,</code> <code>test_AddBribeRewardIsOwnerOnlyAndDelegatesToThePairedBribe ,</code> <code>test_DefaultBoundsAndOwnerAuthorization</code>
S-22	Cumulative reward notifications per token cannot exhaust the reward index	<code>test_LifetimeRewardCapAcceptsTheExactLimitAndRejectsTheFirstExcessUnit ,</code> <code>test_LifetimeRewardCapStillBlocksAfterTheMaximumWasClaimed</code>
S-23	Fund has no administrative surface	<code>test_FundHasNoAdministrativeSurfaceLeft</code>
S-24	Redemption and GBX burning are the only ways assets leave Fund	<code>test_RedemptionIsTheOnlyWayAssetsCanEverLeaveFund</code>
S-25	Reward-token registry is capped at 16 and append-only	<code>test_RewardTokenCountIsPermanentlyCappedAtSixteen</code>
S-26	Signal batches aggregate custody once and roll back every allocation atomically	<code>test_AddSignalManyCustodiesAndMintsAggregateWhileMirroringEveryAllocation ,</code> <code>test_AddSignalManyRollsBackCustodySupplyVotesAndEarlierAllocationWhenLaterAdditionFails ,</code> <code>test_RemoveSignalManyRollsBackEarlierRemovalWhenLaterRemovalFails</code>

32. Liveness properties

ID	PROPERTY	DEPENDS ON
L-1	An account can always remove signal it holds	GBX transferability only
L-2	An account can remove signal, including from a dead Strategy	Reward checkpoints perform no token transfer (§23.8)
L-3	Redemption cannot be paused, gated, or blocked by any party	The redeemer's own token selection
L-4	A redeemer can route around a broken asset by omitting it	Caller-selected basket
L-5	A signaler can claim one reward token while another is frozen	Scalar <code>claimReward(account, token)</code>
L-6	A failed automatic Bribe notification preserves its complete buffer	Atomic <code>BribeRouter.route()</code> ; successful retry may be impossible
L-7	Bribe notification failure cannot revert an already completed Strategy buy	Notification is deferred behind the minimal buffer (§22.4)
L-8	A paid Mine replacement completes without calling downstream Resonance	Successful nominal transfer request to ResonanceRouter (§12.4)
L-9	Signal exit stays available after a Bribe reaches its lifetime reward cap	<code>test_KilledStrategyExitRemainsLiveAfterRewardLifetimeCapIsConsumed</code>
L-10	Every mandatory loop is bounded	<code>MAX_REWARD_TOKENS = 16</code> ; Fund does not loop Mine slots

32.1 Liveness properties that do not hold

NON-PROPERTY	REASON
Rewards in a dead Strategy's Bribe are always claimable	Zero-supply reward time and later notifications can become surplus (§20.4)
Resonance surplus is always recoverable	No recovery path exists (§17.5)
Bribe surplus is always recoverable	No carry, Fund classification, or recovery path exists (§23.4)
A Strategy buy survives Fund transfer failure	Fund receipt is inline and the complete purchase reverts (§22.1)
Router revenue eventually enters the stream	<code>route()</code> needs a transaction and has no guaranteed caller
Accrued mining GBX is always minted	Minting occurs only when the tenure is replaced; effective supply still includes it
Administration is always available	Depends entirely on the unselected external owner (§15, G-03)
An administration call can be observed before it lands	The core provides no delay, queue, or pending state (§29.7)

33. Precision and rounding analysis

33.1 Rounding inventory

#	SITE	FORMULA	DIRECTION	RESIDUE DESTINATION	BOUND PER OPERATION
1	Mine price decay	<code>[P · e/D]</code> subtracted	Favors protocol	none (price is quoted)	≤1 unit
2	Mine payment split	<code>[p · 0.8]</code>	Favors protocol	routed as revenue	≤1 unit
3	Mine next price	<code>[p · m/10¹⁸]</code>	Favors payer	none	≤1 unit
4	New tenure rate	<code>[globalTps/16]</code>	Reduces issuance	unissued forever	<16/s

#	SITE	FORMULA	DIRECTION	RESIDUE DESTINATION	BOUND PER OPERATION
5	Halving rate	$u_0 \gg k$	Reduces issuance	unissued	$\leq 1/s$
6	Resonance schedule rate	$[S/604800]$	Reduces scheduled amount	Resonance surplus	$< 604,800$ raw
7	Resonance global index	$[E \cdot 10^{36}/W]$	Reduces payout	Resonance surplus	denominator-based
8	Resonance per-Strategy	$[w \cdot \Delta/10^{36}]$	Reduces payout	Resonance surplus	< 1 raw/checkpoint
9	Strategy Bribe split	$[p \cdot r/BPS]$	Favors Fund	included in Fund complement	< 1 raw/purchase
10	Bribe schedule rate	$[S/604800]$	Reduces scheduled amount	Bribe surplus	$< 604,800$ raw
11	Bribe global index	$[E \cdot 10^{36}/W]$	Reduces payout	Bribe surplus	denominator-based
12	Bribe account settlement	$[w \cdot \Delta/10^{36}]$	Reduces payout	Bribe surplus	< 1 raw/checkpoint
13	Fund redemption payout	$[bal \cdot g/S]$	Favors remaining holders	stays in Fund	≤ 1 unit/token

Rows 6–8 and 10–12 accumulate unallocated reward surplus with no lifetime bound. Row 9 is intentionally partition-dependent: splitting one economic amount across purchases can direct additional sub-token units to Fund.

33.2 Why 1e36 and not 1e18

With E raw USDG (6 decimals) and W raw sGBX (18 decimals), the relative truncation error of the global index is approximately $1/(E \cdot P / W)$ where P is the precision constant. Taking a representative $W = 10^{22}$ (10,000 sGBX) and one second of a 1 USDG/second stream ($E = 10^6$):

PRECISION P	$[E \cdot P / W]$	RELATIVE ERROR OF ONE CHECKPOINT
10^{18}	$[10^2] = 100$	$\sim 1\%$
10^{36}	$[10^{20}] = 10^{20}$	$\sim 10^{-20}$

At $1e18$, per-checkpoint truncation can be economically material. At $1e36$ it is negligible per checkpoint. Resonance and Bribe both use $1e36$: Bribe reward-token decimals are unconstrained, and a higher-resolution index materially reduces — but does not eliminate — accepted floor surplus.

33.3 The decimal-asymmetry hazard

Six-decimal USDG is a deployment assumption, not a code constant. No contract calls `usdg.decimals()` or validates it. The $1e36$ calibration, every worked example in this document, and every economic figure in the simulation fixtures assume 6 decimals. Binding a USDG with different decimals would leave the contracts functional but invalidate the calibration and all economic modelling. This is discrepancy D-2 (§43).

A related asymmetry applies to Bribe reward tokens, whose decimals are unconstrained. The $1e36$ Bribe index lets low-decimal rewards remain useful across eighteen-decimal signal balances. A raw unit may still be indivisible among accounts; the implementation does not retain account-specific fractions, and the resulting balance is surplus.

33.4 Overflow analysis

EXPRESSION	WIDEST INTERMEDIATE	SAFE BECAUSE
<code>mulDiv(E, 10³⁶, W)</code>	512-bit product	<code>Math.mulDiv</code> full-width intermediate
<code>mulDiv(w, Δ, 10³⁶)</code>	512-bit product	same
<code>mulDiv(bal, g, S)</code>	512-bit product	same
<code>(to - from) · rewardRate</code>	<code>uint256</code>	<code>rewardRate = ⌊S/604800⌋</code> ; <code>S</code> bounded by held balance
<code>mulDiv(emitted, 10³⁶, W)</code>	512-bit product	<code>Math.mulDiv</code> ; lifetime cap bounds cumulative index
<code>elapsed · slot.tps</code>	<code>uint256</code>	<code>tps ≤ 10²⁴</code> ; overflow needs $\sim 10^{52}$ s
<code>mulDiv(balance, Δ, 10³⁶)</code>	512-bit product	<code>Math.mulDiv</code> ; stored index bounded by lifetime cap

The monotonic lifetime cap of §23.6 is checked before notification state or token movement and bounds cumulative index growth (`test_LifetimeRewardCapAcceptsTheExactLimitAndRejectsTheFirstExcessUnit`).

Solidity 0.8.26 provides checked arithmetic throughout; there are no `unchecked` blocks in the value-bearing paths.

34. MEV and timing analysis

34.1 Mining slot auctions

Slot replacement is a public descending-price opportunity with a deterministic, publicly computable price. It is inherently competitive and MEV-exposed:

- **Sniping.** Searchers will fill at the earliest profitable moment. This is the mechanism working as designed — competition compresses the clearing price toward fair value.
- **Front-running a replacement.** Mitigated by `expectedEpochId`: a competing fill increments `epochId` and reverts the victim's transaction rather than executing it at a worse price.
- **Zero-price sniping.** After one hour the price is zero, so a new tenure can begin for free. The new miner may be the outgoing miner; self-replacement is allowed. This is intrinsic to the design and is finding **M-02** (§39.7).

34.2 Strategy auctions

Identical structure. The buyer's protections are `expectedEpochId`, `deadline`, and `maximumPayment`. The auction has no reserve price, so a Strategy's inventory will clear at whatever the market bears, potentially at zero after full decay. Since the next epoch's starting price derives from the last clearing price, a coordinated series of cheap fills can depress subsequent starting prices; the `minimumPrice` floor bounds this, and recovery is geometric at `m_s`.

34.3 Stream restart timing

The sole Router requires its complete balance to be at least both `REWARD_DURATION` raw units and `remainingRevenue`, so an actor wishing to force a Resonance restart must supply the missing amount up to that threshold. Bribe enforces the same two gates directly. The influence that remains is genuine and accepted:

- An actor who *wants* faster emission can top up to restart at a higher rate.
- An actor who *wants* slower emission can top up with `amount ≈ remaining` late in a period, re-spreading the remainder over a fresh seven days.

Both cost real capital proportional to the remainder. There is no free grieving vector, but there is no manipulation-proof guarantee either.

34.4 Signal timing

P-1 (§19) eliminates same-transaction capture. It does **not** eliminate short-horizon strategic signaling: an actor observing an imminent large notification can allocate signal one block earlier and capture that interval's flow legitimately. Because there is no epoch, cooldown, or minimum duration, signal weight is fully fluid. This is a deliberate design decision, not an oversight.

34.5 Redemption timing

A redeemer's payout depends on Mine's effective supply at execution: current `gbx.totalSupply()` after burns plus accrued unminted emission. Adversarial interleaving is bounded:

- Mining accrual is force-checkpointed (§25.3), so it cannot be timed against.
- Unburned Fund GBX inflates the denominator, so a redeemer benefits from burning it first — a permissionless action any redeemer can take in the same transaction bundle.
- Another redemption in the same block reduces both numerator and denominator consistently (I-9), so ordering does not create extractable advantage.

34.6 Checkpoint grieving

Because accrual is lazy and every mandatory loop is bounded, there is no unbounded-work grieving vector. The costs a griever can impose are the fixed 16-token Bribe loop on each Strategy a signaler enters or exits. ADR 0051 multiplies that bounded work across a caller-selected batch; a stale entry may burn gas before atomic rollback. Scalar removal remains the bounded fallback. Fund redemption reads Mine's effective supply in constant time and performs no mining-slot loop.

35. Economic analysis

35.1 The mining market

A miner's expected return over a tenure of length T is:

$$E[\text{return}] = T \cdot \text{tps} \cdot \text{price_GBX} + \Pr[\text{replaced at price } p > 0] \cdot E[0.8p] - p_{\text{entry}}$$

Equilibrium properties:

- **`p_entry` is set by the descending auction**, so it converges toward the market's valuation of $T \cdot \text{tps} \cdot \text{price_GBX}$ plus the option value of a later replacement claim.
- **The replacement-claim term is not guaranteed.** It is zero unless a later replacement clears at a nonzero price, and the price reaches zero after one hour. The later caller may be the same miner.
- **Tenure length is endogenous:** a slot that is profitable to hold is also profitable to take, so T shortens as GBX price rises.
- **The multiplier m is a ratchet.** A hot slot escalates its own next starting price geometrically, damping the rate at which cheap tenures can be acquired in succession.

35.2 Issuance dynamics

The prospective global rate is $u(t)$ per second, halving every 69 days from deployment and reaching u_{∞} at the sixth boundary (§12.5). Two consequences:

- **The tail is the long-run regime.** Almost all of the protocol's lifetime is spent at u_{∞} , not on the halving curve. Parameter selection should therefore weight the tail heavily.

- **Legacy tenures can over-issue relative to the prospective rate** (M-01, §12.6). The excess magnitude is bounded by outgoing tenures' locked rates, but its duration is unbounded because turnover is not guaranteed.

35.3 Signal equilibrium

A signaler's return from allocating weight w to Strategy s over interval $[t_0, t_1]$ is:

Bribe rewards to s over $[t_0, t_1] \cdot w / \text{totalSignalWeight}(s)$

Note what is **absent**: the signaler receives no share of the revenue they direct, and no share of the acquired asset. Revenue directed to s benefits *all GBX holders* via Fund backing, while Bribe rewards accrue only to signalers of s .

This creates the intended separation: **acquisition is a public good funded by protocol revenue; direction is a private good funded by whoever wants that direction**. The equilibrium is that signal flows toward Strategies whose Bribe yield is highest, and Bribe funders bid against each other for the protocol's acquisition capacity.

Failure mode. If no Strategy offers enough reward or other perceived value, signalers may remove their signal, burning their sGBX and recovering the escrowed GBX. Active `totalSignalWeight` can fall toward zero, and stream emission during an interval with zero active signal becomes permanently unclaimable surplus (§17.5). The protocol continues to function but leaks revenue. Nothing in the design guarantees sustained signal demand.

35.4 Redemption and backing

Executable backing per GBX uses `balance_i / effectiveTotalSupply` for each asset i , where effective supply includes accrued unminted mining. It rises when assets are acquired or current GBX supply is burned (Fund burns and redemptions with floored residue). It falls as mining emission accrues. There is no onchain computation of aggregate backing, because that would require prices.

Redemption arbitrage. If the market price of GBX falls below the redeemable value of the basket, burning becomes profitable, which contracts supply and raises backing per remaining token. This is the design's only value-stabilization mechanism, and it is entirely emergent — no contract enforces or subsidizes it.

35.5 What the design does not attempt

The protocol maintains index **membership** — the set of Strategies registered in `Resonance` — but no index **methodology**. There are no target weights, no rebalancing, no drift correction, no diversification constraint, no risk budget, no performance fee, no management fee, no NAV, and no valuation. Fund's composition is a historical record of what signalers directed and buyers filled, plus whatever anyone donated; the proportions are an outcome, not a target.

36. Supported-token model

36.1 Support definition

A Strategy payment token or Bribe reward token is supported only if:

1. each successful transfer moves the requested amount under conventional ERC-20 semantics;
2. `balanceOf`, `approve`, `transfer`, and `transferFrom` behave conventionally;
3. balances do not change asynchronously (no rebasing);
4. the token does not later enable a fee, blocklist, pause, or incompatible upgrade; and
5. callbacks cannot bypass reentrancy guards or authorization.

36.2 Enforcement

ADRs 0047 and 0049 deliberately remove duplicate pre/post balance snapshots from standard-token flows:

BOUNDARY	MECHANISM
Mine payer/Router/claim and SignalGBX GBX custody	SafeERC20 under reviewed canonical USDG/GBX assumptions
ResonanceRouter → Resonance and Resonance → Strategy	SafeERC20 under the standard, non-rebasing USDG assumption
Buyer → Strategy → Fund/BribeRouter	SafeERC20 ; per-purchase arithmetic uses the requested price
Bribe funder/Router → Bribe and Bribe → account	SafeERC20 ; schedule and claim state use the requested amount
Fund → caller-selected redemption receiver	Exact debit/credit checks plus pre-transfer and basket-wide retained-balance guards

The simplified rows do **not** detect every inexact movement. Unsupported mechanics may revert, underfund a schedule, consume a pre-existing donated balance, or short-change a receiver. This is an explicit trust assumption and review obligation, not compatibility.

36.3 Explicit exclusions and accommodations

CATEGORY	HANDLING
Fee-on-transfer	Unsupported ; simplified paths may revert, underfund, or misaccount
Rebasing	Unsupported ; asynchronous balances are outside every stated identity
ERC-777-style callbacks	Authorization and value-bearing entry points are reentrancy-guarded where they carry mutable accounting
Missing-return-value tokens	Accommodated through SafeERC20 when the token otherwise has standard movement semantics
Approval-reset quirks	forceApprove is used at Router notification boundaries
SignalGBX as payment or reward	Forbidden at registration (finding E-03)
Blocklisting or pausable tokens	Unsupported ; may block a purchase, notification, claim, or selected redemption

36.4 Failure isolation summary

FAILURE	BLAST RADIUS
Broken Strategy payment token	That Strategy's purchases and automatic Bribe buffer; Fund transfer failure reverts the buy
Broken Bribe reward token	Scalar claims isolate other tokens; the convenience all-token claim may still revert
Broken Fund asset	Only redemptions that <i>select</i> it
Broken USDG	Systemic — all revenue, all auctions, all mining payments
Broken GBX	Systemic — protocol-controlled, but a core implementation defect is unrecoverable

USDG is the single external token whose failure is systemic and unrecoverable.

37. External dependencies

DEPENDENCY	VERSION/SOURCE	TRUST REQUIRED	FAILURE IMPACT
OpenZeppelin ERC20/Permit/Votes	@openzeppelin/contracts	Library correctness	Systemic
OpenZeppelin SafeERC20, Math, ReentrancyGuard	same	Library correctness	Systemic
USDG	external, third-party	Solvency, no blacklist, no rebase, 6 decimals	Systemic
Strategy payment tokens	external, per Strategy	Standard ERC-20 behavior	Per-Strategy
External UniV2 LP token, if registered	external, one Strategy	Pair, venue, underlying tokens, standard ERC-20	Per-Strategy
Bribe reward tokens	external, per token	Standard ERC-20 behavior	Per-token
EIP-1153 transient storage	chain (Cancun)	tstore / tload availability	Redemption fails

Absent by design: no price oracle, no NAV computation, no entropy source, no keeper network, no cross-chain bridge, no off-chain signer, and no external upgrade authority.

Target chain. README.md names **Robinhood Chain** as the intended target. packages/config/deployments holds dated candidate files. No canonical USDG or bootstrap LP token address is resolved and no signed manifest clears them.

38. Threat model

38.1 Smart-contract bugs

Exposure. The full protocol; nothing is upgradeable. **Mitigation.** Small upstream-shaped reward and settlement surface, immutability, SafeERC20, reentrancy guards, and focused ADR-0048 regressions and mutation coverage recorded in §40. The broader deterministic, fuzz, invariant, and workspace matrix currently recorded there predates ADR 0048 and must be rerun. **Residual.** V12 supplied a finding export for source commit 3ae171b, but it lacks an explicit scope, methodology, named auditor, date, signature, and final assurance statement. Independent disposition accepted the theoretical index-overflow condition, retained a pre-exposure deployment check, and left reward-claim authorization open. A discovered bug cannot be patched, paused, or worked around.

38.2 The Resonance owner (findings G-01, G-03, open)

The core's threat model for administration is exactly one address. Whoever holds Resonance.owner() can add Strategies, kill any Strategy except the final live one, register Bribe reward tokens up to the sixteen-token cap, transfer ownership onward, or renounce it. They **cannot** drain Fund, mint GBX, alter mining economics, reprice outgoing-tenure rates, or operate a core liquidity position — no such surface exists (§9.3). They can move the signaler share within [0, MAX_BRIBE_BPS], which is an economic lever rather than a custody one: it never reaches a completed purchase and never lowers Fund's per-purchase share below 80%. Killing a Strategy is irreversible, making it the highest-impact capture target; renouncing ownership is equally irreversible and permanently freezes Strategy membership.

Because ADR 0034 removed the in-repository Governor and Timelock, the core supplies **no** mitigation of its own: no proposal filter, no quorum, no voting period, no execution delay, no cancellation, and no observable pending state. An owner call takes effect in the transaction that makes it. Every capture, collusion, and liveness question therefore transfers wholesale to the external governance system that has not been selected, and must be re-analyzed against that system's exact release rather than against this repository.

Two properties of sGBX will shape that analysis. First, because no sGBX can be idle (§13.4), `getPastTotalSupply` measures economically active weight only — an external system using it as a quorum denominator is not diluted by parked receipts, and the former undelegated-supply deadlock concern does not arise from the token. Second, and less comfortably, checkpoints survive removal.

38.3 Short-duration voting weight (finding G-01)

sGBX has no signal-removal lock, and its checkpoints record historical block balances permanently. An account may acquire or **borrow** GBX, signal it, allow the snapshot block to pass, and remove immediately afterwards, bearing only borrowing cost and price risk for a few blocks while retaining the recorded weight.

Nothing in the core acts on that weight today, so this is not presently exploitable. It becomes exploitable exactly when an external system is attached that reads historical balances, and its severity then depends on that system's snapshot-to-vote spacing and its proposal threshold. Combined with the irreversibility of `killStrategy`, an inadequately spaced integration would permit a low-cost, permanent hostile action. This is why snapshot compatibility is an explicit item in the ADR 0034 selection requirements (§15.4).

38.4 Oracle and external-price assumptions

None exist. The protocol never reads a price. This eliminates oracle manipulation as an attack class entirely. The substituted risk is that auction clearing prices may be poor if the auction is thin, uncompetitive, or filled at full decay.

38.5 Auction manipulation

- **Depressing Strategy prices.** A coordinated actor filling late repeatedly depresses subsequent starting prices, bounded below by `minimumPrice` with geometric recovery.
- **Inflating Mine prices.** Filling early repeatedly escalates starting prices geometrically at `m`, potentially pricing out honest miners — but each escalation costs the manipulator the full payment, 80% of which goes to the outgoing tenure miner.
- **Inventory timing.** `Strategy.buy` distributes before reading inventory (§19.1), so a buyer always receives everything released through the execution timestamp. They cannot be shortchanged by a stale balance, nor can they acquire inventory that postdates their transaction.

38.6 MEV and transaction ordering

See §34. Mining and Strategy auctions are inherently MEV-exposed; `expectedEpochId`, `deadline`, and price caps convert ordering risk into transaction failure rather than economic loss.

38.7 Reward timing

Stream restarts are influenceable at real capital cost (§34.3). Signal is fully fluid across blocks (§34.4). Neither is manipulation-proof; both are bounded by requiring the manipulator to commit capital proportional to the effect.

38.8 Rounding extraction

Analysis. Resonance and Bribe rate/index/account floors leave value in the reward contract rather than assigning it, so there is no direct recipient to extract. An actor can increase waste through frequent checkpoints at adverse weights, but does not receive that residue. Strategy's per-purchase split floor is different: it assigns the fractional Bribe remainder to Fund through the complement, so payment partitioning can reduce automatic Bribe rewards by sub-token units. ADR 0047 accepts that observable result rather than retaining cross-purchase state.

Verified negative and boundary results: `test_NewStrategySignalCannotReceivePreEntryRoundedSurplus`, `test_ZeroSignalElapsedRevenueBecomesSurplusAndCannotBeCapturedLater`, `test_ElapsedRewardsAtZeroSupplyRemainUnclaimableSurplus`, and `test_EachPurchaseFloorsItsOwnBribeShareWithoutCarry`.

38.9 Malicious and nonstandard tokens

Covered in §36. Blast radius is per-token except for USDG.

38.10 Fee-on-transfer and rebasing assets

Unsupported. The ADR 0047 and ADR 0049 paths intentionally do not prove exact balance deltas, so a fee or rebase can do more than revert: it can underfund a stream or custody position, consume unrelated balance, or short-change a receiver while state advances. The affected market or canonical accounting may become unusable, and there is no adapter or recovery authority. Deployment must review canonical GBX/USDG, and governance must review token code and upgrade controls before registration; those reviews cannot prevent a later proxy upgrade.

38.11 Proxy upgrades in external tokens

Unmitigated and unmitigable. Any Strategy payment token, Bribe reward token, Fund asset, or USDG may be a proxy whose implementation changes to add a blacklist, a fee, a pause, or an arbitrary drain. The protocol has no allow-list to remove them from and no rescue path. A redeemer's only defense is omitting the asset.

38.12 Chain censorship and reorganization

Auction fills, mining replacements, and redemptions are ordinary transactions subject to censorship and reorg. A reorg can undo a fill, changing who holds a slot or who received an auction's inventory. Nothing in the protocol assumes finality beyond ordinary chain guarantees. Prolonged censorship of a slot replacement delays that replacement. Separate censorship or absence of `route()` can leave deposited Mine revenue in ResonanceRouter indefinitely without undoing the completed slot transition.

38.13 Immutable-deployment mistakes

See §28.4. Eight categories of permanent, unrepairable error. This is finding **M-03**, an open High release gate.

38.14 Compromised frontend or indexer

The contracts are permissionless and directly callable, so a compromised interface cannot steal funds directly. It can, however:

- mislead a redeemer into omitting valuable assets (permanently forfeited);
- mislead the final signaler of a dead Strategy into exiting and abandoning rewards;
- present raw balances as inventory, misleading auction participants;
- present the 80% outgoing-tenure replacement claim as guaranteed;
- present unsolicited Fund tokens as protocol-endorsed holdings.

Every one of these is a **permanent** loss to the misled user. Interface correctness is therefore a genuine security boundary, not merely a usability concern.

38.15 Fixed economics pending independent review

Finding **M-04**, open High release gate. Mine's initial rate, provisional 69-day halving period, tail rate, price multiplier, and minimum price are hard-coded and represented in the independent models. They could still produce unsafe or unusable economics despite correct Solidity, so review remains required. The external governance system's own parameters are unelected and outside this repository (§15.4).

38.16 Loss of keys

The deployment coordinator's key is critical **only during setup**, and only until ownership is handed to the external executor (§27.3). Mine and Fund are ownerless, so no key loss affects them. Resonance's owner key is exactly as critical as the external system that holds it, which is unelected; a lost owner key permanently freezes Strategy membership in the same

way `renounceOwnership` would. Loss of a *user's* key loses that user's GBX and any allocated signal permanently, with no recovery mechanism.

38.17 Legal and regulatory risk

Unresolved in every respect. Additionally, upstream code provenance and license reconciliation are explicit release blockers (§41.5): the protocol adapts pinned `give.fun` and Liquid Signal Governance code plus unpinned `donut-miner` lineage, with a transitive **GPL-2.0-or-later** ancestor in the auction lineage and unidentified Synthetix and Solidly ancestors, while the repository declares BUSL-1.1 at the root and MIT per file.

39. Residual risks

Risks that remain after all implemented mitigations, in descending severity.

39.1 External review remains incomplete

V12 supplied 22 Low candidate findings for commit `3ae171b997254b56602298d873b3918d1575b3c7`. The independently reviewed register confirms 249695, 249702, and 249705, rejects 249689's arithmetic premise, and retains the dependency cluster as deployment evidence rather than a canonical runtime exploit. The export is not a complete assurance package or substitute for remediation, retesting, and release review. **Open.**

39.2 Unreparable defects

Immutability means any defect — in code, parameters, or deployment — is permanent. **Accepted by ADR 0016/0017.**

39.3 Unselected external governance

Finding **G-03**. The core has no governance of its own, and the system that will own `Resonance` has not been chosen. Until it is, the protocol's capture resistance, liveness, delay, and accountability properties are undefined rather than weak. A deployment that skipped the handoff would ship an ordinary admin key. **Open release gate.**

39.4 Historical voting weight outlives the position

Finding **G-01**. `sGBX` checkpoints survive removal, so any external system reading historical balances must space its snapshot and voting window deliberately. Not exploitable in the core, which reads no checkpoints. **Open integration gate.**

39.5 Unbounded reward-contract surplus

Schedule-rate, global-index, and account floors, zero-weight intervals, and direct donations accumulate in `Resonance` or `Bribe` with no recovery path and no lifetime bound. **Accepted by ADR 0047.**

Separately, exhaustion of a `Bribe` token's monotonic lifetime cap permanently prevents its paired Router from distributing any later automatic share; the complete Router balance remains stranded because the cap never reopens.

39.6 Unbounded reward abandonment in dead Strategies

A final signaler's exit does not pause the stream. Remaining and later notified rewards can elapse at zero supply and become permanently unallocated. ADR 0028's queue-specific BR-1 model is superseded; the residual risk is **accepted by ADR 0047**.

39.7 Miner rollover and zero-price replacement

Finding **M-02**. The 80% outgoing-tenure replacement claim is contingent, not guaranteed. After one hour the price is zero, and any caller — including the same miner — may replace the tenure without funding a claim. **Accepted by ADR 0024.**

39.8 Transitional over-issuance after a halving

Finding **M-01**. Aggregate issuance can exceed the undivided global rate indefinitely if legacy tenures do not turn over. **Accepted by ADR 0033.**

39.9 Fixed economic parameters pending review

Finding **M-04**. The values are selected and modelled, but independent economic review remains an **open release gate**.

39.10 Lookalike dependencies at deployment

Findings **M-03/E-02**. Reciprocal checks prove consistency, not honesty. **Open release gate.**

39.11 External LP liquidity risk

Registering an external LP token as an ordinary Strategy target creates no liquidity guarantee. The pair, venue, underlying assets, and fungible LP token remain third-party dependencies, and the core has no repair or support path.

39.12 Forfeited redemption assets

A redeemer who omits an asset permanently forfeits their claim to it. Interface error here is unrecoverable.

39.13 Unsolicited Fund assets

Any ERC-20 can become Fund backing without review. Malicious tokens in the Fund cannot harm redeemers who omit them, but can mislead observers about the treasury's composition and quality.

39.14 USDG issuer risk

A single external stablecoin is the substrate for all revenue, all auctions, and all mining payments. Its failure is systemic and unrecoverable.

39.15 Unsupported or mutable token behavior

Governance can register a token that charges fees, rebases, blocklists, pauses, or later upgrades into those semantics; canonical GBX or USDG could also violate its reviewed behavior. ADRs 0047 and 0049 do not verify exact deltas outside Fund redemption, so the impact may be underfunding or misaccounting rather than a clean revert. There is no removal, adapter, or recovery path. **Accepted design constraint; deployment review gate.**

39.16 Legal and provenance

§38.17. **Open release blocker.**

40. Testing and verification evidence

Post-ADR-0050 contract evidence for source **3ae171b** : default Foundry passes 293/293; all 27 invariant entries pass at 1,000 runs of depth 500 with zero handler reverts; integration passes 10/10; and Hardhat passes 4/4, including bytecode parity. Broader workspace, static-analysis, and external-fuzzing reruns remain incomplete. The earlier ADR-0048 and ADR-0047 results below are retained as historical evidence. **None of this constitutes formal proof, an independent audit, deployment approval, or release readiness.**

40.1 Current contract matrix and historical focused evidence

The post-ADR-0050 source at **3ae171b** passed **293/293 default Foundry tests**, all **27 invariant entries** at 1,000 runs of depth 500 with zero handler reverts, **10/10 integration tests**, and **4/4 Hardhat tests**, including bytecode parity. This historical local contract evidence predates and does not cover ADR 0051.

The earlier ADR-0048 source added focused regressions for the sixteen-token Bribe bound and composed SignalGBX moves. Its migration suites passed **104/104**. The named regressions cover permanent rejection of token seventeen, atomic remove-then-add movement, zero/same-Strategy/insufficient-source rejection, complete rollback when the destination addition fails, checkpoint ordering for both Strategies, maximum-bound move gas, and absence of the removed Resonance move selector. The focused mutation campaign kills **47/47** targeted mutants.

Historical maximum-bound gas measurements were: signal addition 491,494; withdrawal 1,129,059; scalar claim 93,018; sixteen sequential scalar claims 1,488,760; all-token claim 1,471,439; Strategy purchase 139,502; composed move with sixteen active streams on both Bribes 1,890,938 against a 3,000,000 ceiling; token-sixteen registration 50,810; and rejected token-seventeen registration 5,349. They do not establish ADR 0051 batch gas.

The immediately preceding ADR-0047 tree passed Foundry **312/312 across 23 suites**, all **29 invariant entries** at 1,000 runs of depth 500 with zero handler reverts, integration **21/21**, Hardhat **4/4**, SDK **47/47**, TypeScript simulations **36/36**, Python environment **5/5**, Python simulations **22/22**, Matchstick **9/9**, web unit **3/3**, Playwright **6/6**, mutation **46/46**, and the listed workspace gates. Those figures predate ADR 0048 and do not constitute a complete current-tree result. The repository-wide format gate remains open because 11 unchanged baseline landing/lockfile files fail Prettier. All results are unpinned local engineering evidence, not independent or release evidence.

40.2 Historical ADR 0044 deterministic workspace matrix

Command: `forge test --summary`

RESULT	VALUE		
Suites	25		
Passed	356		
Failed	0		
Skipped	0		

SUITE	PASSED	SUITE	PASSED
AdversarialTest	18	Superseded liquidity-custody suite	18
ArchitectureReconciliationRegressionTest	4	MineTest	24
BribeBpsTransitionTest	9	ProtocolInvariantsTest	29
BribeRetirementRiskTest	1	ResonanceRouterTest	8
BribeRewardFlowTest	10	ResonanceTest	35
BribeRouterTest	17	SignalGBXTest	23
BribeTest	32	SignalGasTest	4
CarryReallocationTest	4	SixDecimalAutomaticBribeIntegrationTest	1
FactoriesTest	8	SixDecimalBribeInvariantTest	3
FundTest	26	SixDecimalBribeTest	9
GBXTest	10	StartingPointTest	15
HistoricalBribeDifferentialTest	3	StrategyTest	40
USDGFlowTest	5		

This was the ADR 0044 working-tree result before ADR 0047. `ProtocolGovernorTest` and its 11 tests were removed with the Governor itself (ADR 0034). The suites cover the bounded automatic-Bribe rate, six-decimal reward precision,

constant-time Mine accounting, deployment-time halving boundaries, the provisional one-GBX tail, and the ADR 0044 Mine/Router failure-isolation boundary.

Command: `FOUNDRY_PROFILE=integration forge test --summary`

RESULT	VALUE
Suites	2
Passed	19
Failed	0
Skipped	0

`CampaignHarnessTest` 8, `LiquidityFeeHarvestTest` 11.

Historical companion evidence. Hardhat passed 4/4, SDK 50/50, TypeScript simulations 39/39, Python environment-policy checks 5/5 and simulations 25/25, subgraph specification checks 4/4 plus Matchstick 10/10 and build, web unit tests 3/3, Playwright 6/6, and the documentation, ABI, formatting, lint, typecheck, and workspace-build gates. The Forge matrix includes both 10,000-run Mine fuzz cases, 27 stateful invariant entries at 1,000 runs of depth 500 plus two deterministic reachability regressions (29/29 for the suite) with zero handler reverts, and exact gas, harness, fixture, and chart checks.

40.3 Historical campaign configuration

The ADR 0044 run used this `foundry.toml` configuration:

PROFILE	FUZZ RUNS	INVARIANT RUNS	INVARIANT DEPTH	FAIL_ON_REVERT
default	10,000	1,000	500	true
ci	10,000	1,000	500	true
nightly	100,000	10,000	1,000	true
integration	256	—	—	—

Compiler: Solidity 0.8.26, Cancun, optimizer enabled at 10,000 runs, no metadata hash. Foundry 1.7.1 (4072e48705af9d93e3c0f6e29e93b5e9a40caed8).

The `nightly` profile was **not** executed for this document and is not reported as passing.

40.4 Historical evidence by method

Every deterministic method and count in this subsection describes the superseded ADR 0044 working tree, not the later ADR 0047 or current ADR 0048 design.

Unit and negative testing. 356 default-profile tests spanning constructor validation, authorization, degenerate arguments, revert paths, and behavioral regressions for every accepted finding.

Property-based fuzzing. 24 `testFuzz_` properties in the default profile at 10,000 runs each — **240,000 configured fuzz cases**. Properties cover supply reconciliation (I-1), receipt collateralization (I-3), signal-backing (I-4), weighted rate-history frequency-independence (F-21), routing conservation (I-11), redemption pro-rata exactness and monotone backing (I-9), price-curve exactness and monotonicity, next-price bounds, Bribe solvency (I-8), and Resonance non-overpayment (I-7).

Stateful invariant testing. `ProtocolInvariantsTest` contains 27 `invariant_` entries — 26 asserting properties plus `invariant_CallSummary`, which reports selector coverage — and two deterministic regressions (`test_EveryHandlerActionIsReachable`, `test_DynamicallyAddedStrategyEntersEveryHarnessPath`), for the 29 tests reported above. Each invariant entry runs at 1,000 runs × depth 500 with `fail_on_revert = true` — **500,000 calls per entry, 13,500,000 aggregate state-machine transitions**. The reachability regression exists specifically to prevent a permanently short-circuited handler action from producing false confidence; no stale per-selector reach counts are presented as current evidence.

Historical liquidity integration, removed by ADR 0050. The ADR-0044 integration profile contained eleven tests for the then-current canonical-position design. They remain part of the historical count only; the contract and suite no longer exist in the current tree.

Randomized action-sequence campaign. `CampaignHarnessTest` (8 tests) wires the complete protocol graph from a single constructor and runs `testFuzz_RandomActionSequencesPreserveEveryProperty` over 256 randomized 12-action sequences, asserting every property after each action.

Architecture-reconciliation regressions. `ArchitectureReconciliationRegressionTest` (4 tests) is the direct evidence for the ADR 0031/0032 changes: it asserts the removed idle-receipt selectors are absent from **deployed runtime**, that `signal` atomically custodies, mints, delegates, and mirrors the paired Bribe, and that ten one-row-unit payments classify to exactly Fund 9 / Bribe 1.

Differential-model evidence. `packages/simulations` contains independent TypeScript and Python economic models whose committed JSON fixtures and SVG charts are reproducibility evidence, checked by `pnpm simulations:fixtures:check`. The fixtures independently reproduce the mining price curve, the 80/20 mining split, staggered fixed-slot pre/post-halving tenures, the halving schedule, the tail, the Strategy auction curve, the supply identity, and effective-supply redemption dilution.

Gas bounding. `test_MaximumRewardTokenGasStaysFarBelowABlock`, `test_RewardTokenGasSlopeIsRecordedAndBounded`, `test_ScalarSignalEntryAndExitRemainCheapInTheShippedConfiguration`, and `test_FixedLiabilityAndGovernanceGasIsRecorded` bound the mandatory loops of §32 (L-10).

Reward-cap regressions (ADR 0035). `test_LifetimeRewardCapAcceptsTheExactLimitAndRejectsTheFirstExcessUnit`, `test_LifetimeRewardCapStillBlocksAfterTheMaximumWasClaimed`, `test_LifetimeRewardCapFailurePreservesRouterStateAndFundSettlement`, and `test_KilledStrategyExitRemainsLiveAfterRewardLifetimeCapIsConsumed` establish that the monotonic per-token counter is not reset by claims, Fund payment, stream completion, or Strategy death, that a rejected notification leaves the caller's tokens and every liability untouched, and that signal exit stays available after the cap is consumed (§23, L-9).

Checkpoint-retention regressions (ADR 0034). `test_HistoricalVotingCheckpointsSurviveImmediateSignalWithdrawal` and `test_LaterSignalPreservesExplicitDelegateAndSelfDelegatesAgainAfterZeroDelegation` establish the two checkpoint properties §15.3 relies on, and `test_DirectDonationIsSurplusAndCreatesNoSignalVotesOrWithdrawalEntitlement` establishes that a direct GBX donation creates no voting weight.

Static analysis, external fuzzing, and mutation testing — pinned to earlier trees. These campaigns were last executed before the final ADR 0044 Mine/Router boundary; some also predate ADR 0034's Governor removal and ADR 0035's Bribe lifetime cap. They are reported in §40.5 and **must not be cited as current-tree evidence**. Current static-analysis and external-fuzzer reruns remain open; the distinct current focused mutation result is the 47/47 ADR-0048 campaign in §40.1.

Historical companion workspace gates. Hardhat bytecode parity; SDK tests (50/50) and ABI checks; subgraph specification tests (4/4), Matchstick tests (10/10), and build; TypeScript simulations (39/39), Python environment-policy checks (5/5), Python simulations (25/25), and fixture checks; web unit tests (3/3) and Playwright end-to-end tests (6/6); con-

tract and SDK documentation generation; formatting; linting; type checking; and the workspace build all passed on the then-current ADR 0044 working tree.

40.5 Earlier historical evidence — explicitly not current

The former `AUDIT-BASELINE.md` and `TEST-CAMPAIGN.md` records reviewed commit `54e3f2c3ce1de25aea4da2f21-fab27804a3bfa84` (2026-08-09), before the ADR 0024 Mine redesign and the ADR 0029/0030/0031/0032 changes. They were removed from the current audit directory on 2026-08-25 and remain recoverable from Git history. Their reported figures, including **340 default Foundry tests**, describe a superseded contract graph and must not be read as current.

For contrast, the immediately preceding ADR 0042 uncommitted tree also passed **356** default and **19** integration tests. Those counts are retained as dated local engineering history. The later ADR 0044 working tree independently produced the same totals; numerical equality does not make the two runs interchangeable. See discrepancy D-4 for the history of this figure.

Static analysis, external fuzzing, and mutation results are also historical. The pinned pre-ADR-0034/0035 campaign used **Slither 0.11.5**, **Aderyn 0.6.8**, **Semgrep 1.162.0**, **Gitleaks 8.30.1**, with a register of 177 accepted source findings across 28 detector classes and zero raw Semgrep/Gitleaks findings; native **Medusa 1.5.1** at **101,602 calls** with zero failures across 65 surfaces; **Echidna 2.3.2** at **100,213 calls** with all **25 properties** passing; and a focused **43-mutant** campaign that killed every mutant. A later narrow Signal/Resonance mutation rerun killed 49/49 mutants on 21 August 2026 against the ADR 0036/0037 tree. It predates ADR 0044 and is not current Mine evidence. Mythril 0.24.8 was incompatible with constructor-resolved immutable/Cancun runtimes and was never a proof. None of these campaigns covers the complete working tree described by this edition.

40.6 Current verification status and absent methods

METHOD	STATUS IN THE CURRENT DEVELOPMENT TREE
External review	V12 export received for 3ae171b ; independently dispositioned; incomplete and not release-authorizing
Focused migration tests after ADR 0048	104/104 passed (§40.1)
Historical post-ADR-0050 contract matrix	293/293 Foundry; 27 invariant entries; 10/10 integration; 4/4 Hardhat including parity (§40.1)
Complete post-ADR-0051 contract matrix	Open; V12 and the historical counts above do not cover the new API or batch loops
Complete workspace-wide matrix	Incomplete beyond the current contract checks; ADR-0047 workspace counts are historical
Static analysis after ADR 0048	Not re-run (§40.5)
External fuzzing after ADR 0048	Not re-run (§40.5)
Focused mutation testing after ADR 0048	47/47 mutants killed (§40.1)
Repository-wide format gate	Open: 7 unrelated files — 6 landing files plus <code>pnpm-lock.yaml</code> — fail Prettier
Symbolic execution	Not performed (Mythril incompatible with this runtime)
Formal verification	Not performed
Second external-fuzzer seed	Not performed
Current-graph fork execution	Not completed; historical read-only evidence is pinned at Robinhood block 32,035,314 (<code>0xe13569...55aea</code>)
Monitored testnet rehearsal	Not performed
Independent review of selected Mine parameters	Not performed; ADR 0042/0043 constants are modelled but remain provisional
Remaining production/deployment parameters	Not fully selected

METHOD	STATUS IN THE CURRENT DEVELOPMENT TREE
Release review	Not performed
Signed deployment manifest	Does not exist

A skipped fork run is not a pass. Fork results count only when the exact RPC capability and block pin are recorded.

41. Deployment status

41.1 Deployment

The protocol is not deployed on any network. No signed deployment manifest exists for this repository state. `packages/config/deployments` contains dated *candidates* and *policy* files (for example `robinhood-mainnet-wrapped-btc.2026-08-02.candidate.json`, `provisional-mainnet-bytecode-hashes.2026-08-01.json`), none of which is a cleared canonical address.

`docs/DEPLOYMENT.md` is explicitly “an unexecuted development outline, not a deployment manifest or release authorization,” and no script in the repository is authorized to broadcast it.

41.2 Review

Current V12 dispositions are recorded in `packages/contracts/audit/FINDINGS.md` for commit `3ae171b`. The raw export is retained under `packages/contracts/audit/reports`; the prior pre-ADR-0047 campaign ledger was removed from the current audit directory and remains available through Git history.

41.3 Open release gates

FINDING	SEVERITY	GATE
M-03	High	Immutable bindings cannot detect a malicious lookalike. Requires signed manifest, exact runtime code hashes, constructor arguments, and receipts.
M-04	High	Mine economics are selected, hard-coded, and modelled, but still require independent economic review before deployment.
G-03	High	The external governance system that will own <code>Resonance</code> is unselected; its voting, delegation, permission, and delay semantics are unreviewed.
G-01	High	sGBX checkpoints survive removal; the selected external system’s snapshot-to-vote spacing requires independent review of the capture model.
E-02	High	Materially reduced by binding checks and post-deployment Mine/Router verification, but codehash, parameter, and manifest review remains external.
V12-249702	Low	An empty Mine slot can be captured before the GBX handoff; inspect every slot and abandon/redeploy a touched candidate before exposure.
V12-249705	Low	Permissionless claims can force another account’s sub-unit Bribe floors; claim authorization remains open and no remediation has been selected.

41.4 Accepted findings (not gates)

A-02 (reward-floor surplus) remains accepted under ADR 0047. ADR 0027’s **A-09** carry-classification treatment and ADR 0028’s queue-created **BR-1** terminal state are superseded; §20.4 records the remaining zero-supply abandonment risk. **M-01**, **M-02** remain accepted by ADR 0024. **G-02** was superseded by governance removal. **A-08** retains its bounded sixteen-token loop under ADR 0048. **SR-001** was fixed by ADR 0031. **SR-002**’s 100%-Fund rule was fixed by ADR 0032, but its

later frequency-independent split machinery is intentionally removed by ADR 0047 (§22.3). **V12-249695** is an accepted theoretical risk under the canonical six-decimal USDG identity and supply assumption; no source change was selected.

41.5 Legal and provenance

An **unresolved release blocker**. `docs/LEGAL-PROVENANCE-BLOCKER.md` records that the active contracts adapt pinned `give.fun ef6ee14a...`, Liquid Signal Governance `14b5fbbb...`, and unpinned donut-miner lineage; that `Strategy`'s descending-price shape has a transitive Euler Fee Flow ancestor at `3bee858a...` whose reviewed file is **GPL-2.0-or-later**; that stated Synthetix and Solidly ancestors lack exact repository, commit, and path; and that the repository declares BUSL-1.1 at the root while every active Solidity file declares MIT. No clean-room, compatibility, relicensing, or separate-permission claim is made.

41.6 Status language

The following terms are **not** applicable as release or security status for this protocol in this development tree: *audited, safe, verified, launched, live, production-ready, trustless, risk-free, fully decentralized, community-owned, guaranteed yield*.

Terms that **are** supported by evidence: *immutable* (no upgrade path exists in `packages/contracts/src`), *governance-minimized* (four selector-bounded continuing actions), *ownerless* (of `Mine` and `Fund`), *permissionless* (of the named user-facing operations), and *non-transferable* (of `sGBX`).

42. Contract reference

CONTRACT	PATH (UNDER <code>PACKAGES/CONTRACTS/SRC</code>)	INHERITS	KEY CONSTANTS
<code>GBX</code>	<code>core/GBX.sol</code>	<code>ERC20</code> , <code>ERC20Permit</code>	Starts at zero supply
<code>Mine</code>	<code>core/Mine.sol</code>	<code>ReentrancyGuard</code>	<code>PRICE_MULTIPLIER</code> 2, <code>MIN_INITIAL_PRICE</code> 1e6, <code>INITIAL_TPS</code> 64 ether, <code>HALVING_PERIOD</code> 69 days (provisional), <code>TAIL_TPS</code> 1 ether, <code>SLOT_COUNT</code> 16
<code>SignalGBX</code>	<code>core/SignalGBX.sol</code>	<code>ERC20</code> , <code>ERC20Votes</code> , <code>ReentrancyGuard</code> , <code>Ownable</code>	—
<code>Resonance</code>	<code>core/Resonance.sol</code>	<code>ReentrancyGuard</code> , <code>Ownable</code>	<code>REWARD_DURATION</code> 7 days, <code>REWARD_PRECISION</code> 1e36, <code>MAX_BRIBE_BPS</code> 2_000
<code>ResonanceRouter</code>	<code>core/ResonanceRouter.sol</code>	<code>IResonanceRouter</code> , <code>ReentrancyGuard</code>	Immutable USDG and Resonance endpoints
<code>Strategy</code>	<code>core/Strategy.sol</code>	<code>ReentrancyGuard</code>	<code>BPS</code> 10_000, <code>MIN_EPOCH_DURATION</code> 1 hours, <code>MAX_EPOCH_DURATION</code> 365 days, <code>PRICE_SCALE</code> 1e18, <code>ABSOLUTE_MINIMUM_PRICE</code> 1e6
<code>StrategyFactory</code>	<code>core/StrategyFactory.sol</code>	<code>Ownable</code>	—
<code>Bribe</code>	<code>core/Bribe.sol</code>	<code>ReentrancyGuard</code>	<code>REWARD_DURATION</code> 7 days, <code>REWARD_PRECISION</code> 1e36, <code>MAX_REWARD_TOKENS</code> 16, <code>MAX_LIFETIME_REWARD_AMOUNT</code> $[(2^{2^5 6}-1)/1e36]$
<code>BribeFactory</code>	<code>core/BribeFactory.sol</code>	<code>Ownable</code>	—
<code>BribeRouter</code>	<code>core/BribeRouter.sol</code>	—	Immutable payment token and paired Bribe; thresholded complete-balance <code>route()</code>
<code>Fund</code>	<code>core/Fund.sol</code>	<code>ReentrancyGuard</code>	<code>REDEMPTION_NAMESPACE</code>

Interfaces: `IResonance`, `IResonanceIdentity`, `IBribe`, `IMine`, and `IResonanceRouter`. `ISignalGBXAllocation` was **deleted** by ADR 0031.

There are 11 core contract types and five core interfaces. Brittle source line counts are intentionally omitted. There is no governance/ source tree: ADR 0034 deleted `ProtocolGovernor.sol` along with its tests, ABIs, SDK lifecycle helpers, and subgraph data sources.

42.1 Permissionless entry points

`Mine.mine`, `Mine.claimMinerPayment`; `ResonanceRouter.route`; `Resonance.distributeRevenue`; `SignalGBX.addSignal` / `addSignalMany` / `removeSignal` / `removeSignalMany`; `Strategy.buy`; `Bribe.notifyReward` / `claimReward` / `claimRewards`; `BribeRouter.route`; `Fund.burnGBX` / `redeem`; `GBX.burn`. `SignalGBX.delegate` / `delegateBySig` are permissionless but read by nothing in the core (§15.3).

42.2 Restricted entry points

FUNCTION	CALLER
<code>GBX.mint</code>	<code>Mine</code> (locked)
<code>GBX.setMinter</code>	current minter, once
<code>Resonance.addSignalFor</code> / <code>removeSignalFor</code>	<code>SignalGBX</code>
<code>Resonance.notifyRevenue</code>	<code>ResonanceRouter</code>
<code>Resonance.addStrategy</code> / <code>killStrategy</code> / <code>addBribeRewardToken</code> / <code>setBribeBps</code>	owner (unselected)
<code>Resonance.setResonanceRouter</code>	owner, once
<code>Bribe.addSignalWeight</code> / <code>removeSignalWeight</code> / <code>addRewardToken</code>	<code>Resonance</code>
<code>StrategyFactory.createStrategy</code> , <code>BribeFactory.createBribe</code>	bound <code>Resonance</code>
<code>SignalGBX.setResonance</code> , factories' <code>setResonance</code>	owner, once
<code>transferOwnership</code> / <code>renounceOwnership</code> on each Ownable contract	its current owner

43. References and provenance

43.1 Source of truth

ARTIFACT	PATH
Solidity implementation	<code>packages/contracts/src</code>
Executable specification suite	<code>packages/contracts/test/minimal/StartingPoint.t.sol</code>
Internal fact registry	<code>docs/facts/gumball-6900-facts.md</code>
Finding register	<code>packages/contracts/audit/FINDINGS.md</code>
Economic fixtures	<code>packages/simulations/fixtures/economic-scenarios.json</code>

43.2 Authoritative ADRs

ADR 0017 (ownerless Fund and no successor; liquidity clauses superseded by ADR 0050) · ADR 0024 (immutable multi-slot Mine; its GBX-ERC20Votes statement superseded by ADR 0030 and synchronous downstream route by ADR 0044) · ADR 0029 (Bribe-based Resonance; signal entrypoints superseded by 0030 then 0031, kill-final-Strategy by 0031, 100%-Fund by 0032, intended Timelock owner by 0034, Mine call-site behavior by 0044) · ADR 0030 (non-transferable ERC20Votes sGBX; its `ProtocolGovernor`, Timelock, selector-filter, and cancellation decisions superseded by ADR 0034,

and its idle-sGBX and `allocatedBalance` decisions by ADR 0031, and its dedicated Resonance move hook by ADR 0048) · ADR 0031 (mandatory signal-backed SignalGBX) · ADR 0032 (acquired-asset Fund/Bribe split; deferred settlement superseded by ADR 0047) · ADR 0033 (fixed sixteen Mine slots and constant-time pending emission) · ADR 0034 (external governance ownership) · ADR 0035 (Bribe lifetime reward cap) · ADR 0036 (governed global Bribe share; exact split carry superseded by ADR 0047) · ADR 0037 (high-precision Bribe index) · ADR 0038 (fixed Mine economics) · ADR 0039 (event-only Mine messages) · ADR 0040 (deployment-time Mine authority verification) · ADR 0041 (time-based Mine halvings) · ADR 0042 (provisional accelerated Mine emissions) · ADR 0043 (provisional one-GBX tail) · ADR 0044 (Mine deposits without synchronous revenue routing) · ADR 0045 (post-deployment Mine/Router token verification) · ADR 0046 (USDG-only scalar Resonance; exact schedule mechanics superseded by ADR 0047) · ADR 0047 (Synthetix-shaped reward streams and direct Strategy settlement; preserved eight-token cap superseded by ADR 0048) · ADR 0048 (fixed sixteen-token Bribe bound and composed SignalGBX moves) · ADR 0049 (canonical GBX/USDG SafeERC20 transfers without balance-delta checks; Fund redemption checks preserved) · ADR 0050 (zero premint and an external LP token treated as an ordinary Strategy asset, with no liquidity-specific core mechanism).

43.3 Superseded ADRs excluded from this document

Fully superseded: ADR 0018 (auto-compounding LP → ADR 0022) · ADR 0022 (fixed-principal LP fee routing → ADR 0050) · ADR 0021 (uniform 100%-Fund Strategy settlement → ADR 0032) · ADR 0023 (fixed supply and Fundraiser reserve → ADR 0024) · ADR 0025 (global revenue stream → ADR 0026) · ADR 0026 (exact successor stream → ADR 0029, then ADR 0047) · ADR 0027 (Bribe carry boundaries → ADR 0047) · ADR 0028's queue-and-pause terminal design (→ ADR 0047).

Partially superseded, historical context only: ADR 0013 (acquisition splits, buyback, proposer model) · ADR 0014 (mint authority, distribution, fee routing) · ADR 0015 (whole-account actions, public coordination surface) · ADR 0016 (terminology; “management fee” means the bounded acquisition auction; exact reward-token cap → ADR 0048) · ADR 0019 (Resonance batch APIs, direct signal entrypoints, idle allocation, standalone exit, and exact eight-token cap) · ADR 0020 (reward carry, donation synchronization, and Strategy routing — superseded across ADRs 0021, 0027, 0032, and finally 0047).

43.4 Upstream lineage

Pinned `give.fun` `ef6ee14a454432210d13e312d0ef825f670bd79d`; Liquid Signal Governance `14b5f-bbbe1945f2e6501f84976e5f12b39fb227a`; donut-miner exact revision unresolved; transitive Euler Fee Flow `3bee858a1568d1313f37d615953f83391a897866` (GPL-2.0-or-later); stated Synthetix `StakingRewards` and Solidly ancestors, **exact sources unresolved**. Full per-file mapping and SHA-256 evidence: NOTICE and docs/LEGAL-PROVENANCE-BLOCKER.md. See §41.5 — this is an open release blocker.

43.5 Recorded discrepancies

D-1 — “Reverse Dutch auction” naming. `AGENTS.md`, `docs/EMISSIONS.md`, and the contracts' own `NatSpec` use “reverse Dutch.” Mechanically both `Mine` and `Strategy` implement a **descending-price** auction, conventionally a plain Dutch auction; “reverse Dutch” follows the Euler Fee Flow lineage. *Resolution:* the repository's term is retained and the mechanics are always stated explicitly.

D-2 — Six-decimal USDG is an assumption, not a constant. No contract reads or validates `usdg.decimals()`. The only in-repository evidence for 6 is the test fixture (`ProtocolFixture.sol:65`, `MockERC20("Global Dollar", "USDG", 6)`) and the simulation fixtures. *Resolution:* stated throughout as a deployment property of the intended USDG, never as a contract guarantee. See §33.3.

D-3 — Historical Fundraiser design. The design was superseded by ADR 0024, and no `Fundraiser.sol` exists in either the current source tree or generated artifact trees. *Resolution:* retain references only where explicitly labeled as historical context; never present a Fundraiser as part of the current protocol.

D-4 — Test-count drift. The deleted historical `TEST-CAMPAIGN.md` reported 340 at commit `54e3f2c3` (2026-08-09); the earlier finding register reported 322 at `281e601` while the actual figure there was 339; the pre-ADR-0034 campaign recorded 335 default and 17 integration; and the later `dc67d7c` tree recorded 329 default and 18 integration. The immediately preceding ADR 0042 tree recorded 356 default and 19 integration; the later ADR 0044 working tree independently passed the same totals. *Resolution:* all of those counts are historical. The later ADR-0047 tree separately passed 312/312 Foundry tests across 23 suites, all 29 invariant entries at 1,000 runs of depth 500 with zero handler reverts, and 21/21 integration tests; those figures also predate ADR 0048. Focused ADR-0048 evidence is 104/104 migration tests and 47/47 killed targeted mutants. The post-ADR-0050 contract source at `3ae171b` passes 293/293 default Foundry, all 27 invariant entries at 1,000 runs of depth 500 with zero handler reverts, 10/10 integration, and 4/4 Hardhat including parity. The broader workspace matrix remains incomplete. These are unpinned development results, and the repository-wide format gate remains open because 7 unrelated files — six landing files plus `pnpm-lock.yaml` — fail Prettier.

D-5 — Ownership closure is procedural. `docs/ACCESS_CONTROL.md`, `docs/INVARIANTS.md`, and `docs/TRUST_ASSUMPTIONS.md` state ownership and administrator conditions as invariants. **No Solidity enforces them;** they are deployment steps 9–10. Under ADR 0034 the specific obligation changed — from Timelock role closure to transferring Resonance to a reviewed external executor and proving the setup owner retains nothing — but its procedural character did not. *Resolution:* described throughout as deployment obligations requiring signed evidence. See §27.2 and §27.3.

D-7 — Documents were re-derived after a mid-work protocol change. An earlier version of this whitepaper, the one-pager, the article, and the fact registry described commit `281e601`. That commit was superseded by ADR 0031 (mandatory signal-backed SignalGBX) and ADR 0032 (fixed 90/10 acquired-asset settlement) before publication. The superseded drafts asserted that 100% of every auction payment became a Fund liability, that no auction proceeds fund Bribes, that idle sGBX could vote without earning, and that a standalone staking surface existed — **all four are now false**. *Resolution:* every affected claim was re-derived against `95ed60e` and the suites re-run. Any copy of these documents citing `281e601` should be discarded.

D-8 — Governance was removed after the previous edition was written. Edition 1.x of this whitepaper, the one-pager, the article, and the fact registry documented an in-repository `ProtocolGovernor` and `TimelockController`, including quorum formula F-8, the selector-bounded proposal filter, disabled-surface behavior, and the uncancellable queued-operation finding G-02. ADR 0034 deleted both contracts. Those sections previously carried “superseded” banners while retaining present-tense claims; in this edition they are **replaced** by §15 and §27, which specify the owner-gated surface that actually exists and state explicitly which guarantees no longer have any source of enforcement. Findings **G-02** and the local-parameter form of **G-03** are superseded by removal, not proven safe. Any copy of these documents describing a Governor or Timelock as current is stale.

D-9 — Reward and Strategy complexity was removed after the previous edition. The ADR 0046 edition described front-loaded exact raw-unit schedules, Bribe queues and zero-supply pauses, precision carry classified to Fund, caller-selected batch claims, exact balance-delta enforcement, and BribeRouter custody of complete Strategy payments through dual liabilities. ADR 0047 intentionally superseded those mechanisms in favor of scalar Synthetix state, ordinary leftover rollover, accepted floor surplus, per-purchase direct Fund settlement, and a Bribe-only buffer. *Resolution:* §§16–23 and §§29–40 are re-derived from ADR 0047; older copies remain historical only.

D-10 — Canonical transfer-delta checks were removed after the previous edition. ADR 0049 removes the repeated sender/receiver balance snapshots from Mine and SignalGBX. These paths now use `SafeERC20` and trust the reviewed canonical GBX/USDG implementations, consistently with the standard-token model. Fund keeps its exact selected-token payout and basket guards because arbitrary addresses can enter its redemption list. *Resolution:* §§12–14, §36, and §§38–40 describe the new boundary; earlier exact-delta descriptions remain historical.

D-11 — Canonical genesis liquidity was removed. ADR 0050 removes the 20 million GBX premint and the canonical liquidity contract. GBX now begins at zero supply and Mine is the sole lifetime issuer. A reviewed external fungible USDG-GBX UniV2 LP token may be registered only as an ordinary bootstrap Strategy asset. *Resolution:* §§11, 26, 28, 35–42 and the economic fixtures use the zero-offset model and contain no liquidity-specific core mechanism.

D-6 — “Index protocol” framing. `README.md` calls the protocol an “onchain index protocol,” while `AGENTS.md:72` uses the narrower and more precise formulation: “Official protocol/index membership is represented by Strategies registered in Resonance, not by a Fund asset list.” *Resolution:* this document adopts the `AGENTS.md` sense throughout. Registered Strategies **are** index membership — the curated list of target assets — but the protocol supplies no index methodology: no weights, rebalancing, drift correction, reconstitution, or NAV (§4 N1, §35.5). Membership is never inferred from a Fund balance, since Fund accepts unsolicited transfers without review (§24.2).

43.6 Companion documents

- [GumBall6900 at a Glance](#) — one-page summary
- [How GumBall6900 Turns Community Conviction Into an Onchain Portfolio](#) — plain-English explanation
- [Internal fact registry](#) — per-claim evidence with source, tests, and caveats

Current source is an uncommitted post-ADR-0051 development tree based on V12-reviewed commit `3ae171b997254b56602298d873b3918d1575b3c7`. The later delta is not covered by V12. Not deployed or approved for user funds.